



Research Paper

Investigating the optimal number of topics by advanced text-mining techniques: Sustainable energy research

Amer Farea^{a,b}, Shailesh Tripathi^{a,d}, Galina Glazko^c, Frank Emmert-Streib^{a,*}^a Predictive Society and Data Analytics Lab, Faculty of Information Technology and Communication Sciences, Tampere University, Tampere, 33720, Finland^b Faculty of Engineering and Information Technology, Taiz University, Taiz, Yemen^c Department of Biomedical Informatics, University of Arkansas for Medical Sciences, Little Rock, USA^d Josef Ressel Centre for Data-Driven Business Model Innovation, University of Applied Sciences Upper Austria, Wehrgrabengasse 1-3, 4400 Steyr,

ARTICLE INFO

Keywords:

Natural language processing
Topic modeling
Latent Dirichlet Allocation
Transformer model
Sustainable energy

ABSTRACT

In recent years, there has been a growing interest in analyzing text data from different scientific fields. The significant advancement of Artificial Intelligence in Natural Language Processing enables a systematic categorization of the wealth of scientific papers into fundamental thematic clusters. In this context, topic modeling is playing a crucial role. Unfortunately, the comparative analysis between traditional and advanced topic modeling methods, including well-established techniques like Latent Dirichlet Allocation (LDA) and newer approaches like BERTopic, remains significantly underexplored. This study addresses this gap by conducting a comprehensive analysis of extensive text data focused on sustainable energy research. To achieve this, we compile a unique dataset consisting of thousands of abstracts sourced from PubMed, Scopus, and Web of Science. Our analysis involves a comparison between LDA and the transformer model BERTopic. Importantly, we introduce a novel approach to determine the optimal number of topics, achieved through the maximization of combined semantic scores, and show that the number of topics is considerably lower than from previous approaches. Overall, our study not only contributes methodologically but also enhances our understanding of the principal topics in sustainable energy research.

1. Introduction

Topic modeling plays a crucial role in managing vast amounts of textual data by aiding in improved information retrieval, content summarization, and understanding of context. As text data complexity and volume continue to rise, topic modeling remains indispensable for extracting valuable insights from unstructured text. This makes it an important field of research within the domains of natural language processing and machine learning.

Leveraging advancements in machine learning and statistical research, novel techniques have emerged for uncovering word patterns within document compilations through hierarchical probabilistic models. These models, referred to as “topic models”, facilitate the discovery of underlying themes that shape the content of documents. Notably, topic models have demonstrated utility in analyzing diverse content beyond just textual information, including imagery, biological data, and survey responses (Blei and Lafferty, 2006; Srivastava and Sahami, 2009). An important application of topic modeling is to uncover research trends (Sharma et al., 2022; Yu and Xiang, 2023).

The primary significance of topic modeling lies in its capacity to unveil patterns in word usage and establish connections among documents that exhibit comparable patterns. The essence of topic models revolves around terms that collaboratively interact with documents, resulting in documents being amalgamations of various topics. In this context, a topic represents a probability distribution across words. In simpler terms, a topic model serves as a generative framework for documents. It outlines a straightforward probabilistic process through which documents can be synthesized.

The process of modeling a document begins with the initial selection of a distribution across topics. Then, each word in the document is probabilistically allocated to a topic according to this distribution. Finally, a word is chosen from the topic to which it has been assigned (Steyvers and Griffiths, 2007).

In the area of text analysis, topic models usually rely on the “bag-of-words” approach, discarding word order. Each document in a corpus is represented by a histogram of word occurrences. This histogram is governed by a distribution over a specific number of topics, with each

* Corresponding author.

E-mail addresses: amer.farea@tuni.fi (A. Farea), shailesh.tripathi@fh-steyr.at (S. Tripathi), GVGlazko@uams.edu (G. Glazko), frank.emmert-streib@tuni.fi (F. Emmert-Streib).<https://doi.org/10.1016/j.engappai.2024.108877>

Received 21 February 2024; Received in revised form 31 May 2024; Accepted 21 June 2024

Available online 8 July 2024

0952-1976/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

topic having a distribution over words in the vocabulary. Learning these distributions results in a compressed representation of high-dimensional histograms for each document, capturing their essential characteristics (Hofmann, 1999, 2001). Over the years, a number of different topic modeling techniques have been introduced, including Non-negative Matrix Factorization (NMF) (Lee and Seung, 1999), Probabilistic Latent Semantic Analysis (PLSA) (Hofmann, 1999, 2001), Latent Dirichlet Allocation (LDA) (Blei et al., 2003), Hierarchical Latent Dirichlet Allocation (hLDA) (Griffiths et al., 2003), Dynamic Topic Models (DTM) (Blei and Lafferty, 2006; Srivastava and Sahami, 2009), Correlated Topic Model (CTM) (Blei and Lafferty, 2007).

Over time, topics within a document corpus can evolve. Ignoring temporal aspects during topic modeling can obscure topic discovery. To address this, topic evolution modeling incorporates time, revealing concealed insights within the corpus. It identifies topics influenced by temporal patterns, enabling an examination of their evolution. Various domains find utility in topic evolution models, such as researchers studying the progression of research topics and identifying pertinent documents.

While conventional topic modeling methods treat documents as collections of words, neglecting the nuances of word meanings and the context within sentences, text embedding techniques avoid this limitation. Examples for text embedding approaches are Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2018) and its variations (Lee et al., 2020; Liu et al., 2019; Lan et al., 2019). Such approaches have been also utilized for topic modeling. Examples include Top2Vec, which utilizes Doc2Vec's word and document representations to jointly embed topic, document, and word vectors (Angelov, 2020; Le and Mikolov, 2014). Furthermore, HDBSCAN (McInnes and Healy, 2017; Sia et al., 2020) employ cluster embeddings to represent topics, and they extract topic representations by identifying words in close proximity to the center of a cluster. While this idea is interesting, it has problems when clusters are not organized spherically. To improve upon this shortcoming (Sia et al., 2020) attempts to address this by re-ranking topic words based on their frequency within a cluster.

Numerous studies have emphasized the effectiveness of employing Latent Dirichlet Allocation (LDA) in diverse communicative contexts, particularly with short texts (Hall et al., 2008; Hong and Davison, 2010; Tong and Zhang, 2016; Maier et al., 2021). Despite evidence supporting the effectiveness of topic modeling with short texts (Sbalchiero, 2018), there remains a lack of empirical support when it comes to its application on longer texts.

While some studies propose a basic guideline of segmenting input texts into 1000-word chunks (Jockers and Mimmo, 2013), there is a notable absence of systematic investigations validating such assertions. Indeed, this limitation of topic modeling extends to various other text analysis techniques. Whereas short texts – such as abstracts, social media posts, journal articles, newspaper articles, and Wikipedia entries – tend to succinctly encapsulate main themes, the application of text mining to longer texts reveals lower content density (Michel et al., 2011). A study by Sbalchiero and Eder (2020) attempts to illuminate the relationship between text length and the optimal number of topics.

Importantly, determining the optimal number of topics to detect within a corpus significantly impacts the outcome. The parameter ' k ' in the algorithm holds particular importance as it influences the fitting model process, thereby affecting the validity of the results. If the number of topics is too small, it may result in broad and heterogeneous topics, while a large ' k ' value may lead to overly specific topics, both of which pose challenges for interpretation. Various approaches have been proposed to address this issue (Cao et al., 2009; Arun et al., 2010; Deveaud et al., 2014), with one of the most straightforward being a Bayesian method. This method involves calculating the log-likelihood for all possible models within a specified range (e.g., 2 to N) using the Gibbs sampling algorithm to derive samples from the posterior distribution of topics within documents, thus identifying the optimal number of topics for the model (Griffiths and Steyvers, 2004). To assess

the implications of different numbers of topics (T_j , where $j = 1, \dots, n$), the authors suggest computing the posterior probability of these models based on the observed data (words in the corpus). The likelihood of the data given the number of topics ($P(w|T)$) is evaluated, indicating the extent to which the latent topic structure produces documents reflecting the observed data, i.e., the corpus. It is intuitive to understand that $P(w|T)$ increases to a peak and then decreases thereafter, with the maximum value indicating the optimal number of topics to fit the model. Despite considerable efforts in addressing the challenge of pinpointing the optimal number of topics, a definitive solution remains elusive, presenting a persistent obstacle.

In this study, we evaluate the performance of LDA (Blei et al., 2003) and BERTopic (Grootendorst, 2022) in topic modeling with text data from sustainable energy research. Although both models are popular, neither inherently includes a mechanism for determining the optimal number of topics, as they operate in an unsupervised manner. Instead, for both models the number of topics is identified either from a clustering of topics or the optimization of external measures. As a consequence, many studies overestimate the number of topics for LDA and BERTopic which leads to considerable challenges when interpreting the obtained topics. Our main contribution is the introduction of a method for determining the optimal number of topics based on normalized scores, including perplexity and coherence scores, providing a consolidated evaluation of topic quality. To achieve this, we frame the issue as an optimization problem. We study this approach for time independent as well as time dependent text data from sustainable energy research and show that the latter allows to gain insights into the evolution of the field over time.

This paper is organized in the following way. In the next section, we discuss related work followed by a discussion of our research questions. In Section 4, we introduce the data and methods we use for our analysis and in Section 5 we present our results. In Section 6 we provide a discussion of our results and in Section 7 the paper finishes with concluding remarks.

2. Related work

Latent Semantic Analysis (LSA) is a Natural Language Processing (NLP) technique that aims to create vector-based representations of texts to capture their semantic content (Dumais et al., 1994; Landauer and Dumais, 1997; Landauer et al., 1998). LSA estimates text similarities by generating vector representations, enabling the identification of related words efficiently. Originally called Latent Semantic Indexing (LSI), LSA has evolved to enhance information retrieval tasks, aiding in selecting relevant documents from a large pool based on their proximity to a query. LSA encompasses various approaches, including keyword matching, weighted keyword matching, and vector representation based on word occurrences in documents. This method employs Singular Value Decomposition (SVD) to structure the data effectively. Specifically, SVD is a technique that utilizes matrices to restructure and compute different dimensions within a vector space by arranging dimensions in order of significance, from the most important to the least important one.

In general, Latent Semantic Analysis (LSA) is used to extract the meaning of the text, focusing on the most significant assumptions while disregarding less important ones. Words with similar vectors will exhibit a high degree of semantic similarity. The key steps in LSA involve the gathering of relevant text data, segmenting it into documents, and creating a co-occurrence matrix for terms and documents, where cells are labeled by document x , term y , and values m for term dimension and n for document dimension.

Probabilistic Latent Semantic Analysis (PLSA) was introduced by Hofmann (1999, 2001) as an enhancement over the LSA method. PLSA addresses LSA's limitations by automating document indexing through a statistical latent class model for factor analysis of count data. It leverages a generative model to probabilistically improve upon

LSA. The primary aim of PLSA is to differentiate word usage contexts without relying on a dictionary, with two key benefits: disambiguating words with multiple meanings and revealing similarities by grouping words that share a common context.

PLSA uses a latent variable $z_k \in \{z_1, z_2, \dots, z_K\}$ to represent semantic layers, forming the foundation of its algorithmic structure. The complete model, denoted as $p(d_i)$ for document probability in the dataset, involves $p(w_j|z_k)$ representing semantics through z_k -related terms (words), and $p(z_k|d_i)$ indicating semantic document distribution. Using these components, the model generates new data via a series of steps (Hofmann, 1999, 2001).

PLSA is presented through two distinct formulations. The first, the symmetric formulation, involves deriving words (w) and documents (d) from the latent class (c) using conditional probabilities $P(d|c)$ and $P(w|c)$ in similar ways. The second, the asymmetric formulation, selects a latent class for each document (d) based on $P(c|d)$ and generates words from that class according to $P(w|c)$. These formulations have been enhanced with the release of Recursive Probabilistic Latent Semantic Analysis (RPLSA), which improves both asymmetric and symmetric variants (Bassiou and Kotropoulos, 2011).

PLSA finds applications in diverse fields like information retrieval, natural language processing, and machine learning from text. It serves tasks such as automatic essay grading (Srinivasarao and Sharaff, 2022; Kakkonen et al., 2008), classification (Chandran et al., 2023; Bosch et al., 2006), topic tracking (Yu and Xiang, 2023; Watanabe et al., 2011; Alghamdi and Alfalqi, 2015), image retrieval (Shamna et al., 2022; Romberg et al., 2009), and automatic question recommendation (Li et al., 2022; Wu et al., 2008).

The Latent Dirichlet Allocation (LDA) model (Blei et al., 2003) was introduced to enhance mixture models, improving the way they capture the exchangeability of words and documents compared to PLSA and LSA. This approach is based on the classic representation theorem stating exchangeable random variables can be represented by infinite mixtures (Blei et al., 2003).

The proliferation of electronic document collections, including web content, blogs, news, and literature, has posed challenges, spurring the need for automated methods to analyze, summarize, and visualize these collections. Latent topic modeling, like LDA (Ramathulasi and Rajasekharababu, 2022; Nallapati et al., 2008), has gained popularity as an unsupervised technique for discovering topics within large document sets. In essence, LDA represents each document as a mixture of topics, where each topic is a discrete probability distribution indicating the word likelihood within that topic. These topic probabilities succinctly capture document content, treating documents as “bags of words” (BoW) without structural hierarchy. Diverse LDA-based variations have been developed, including applications in temporal text mining (Wu et al., 2023; Alghamdi and Alfalqi, 2015), author-topic analysis (Zhang and Lauw, 2022; Ho et al., 2018; Liu et al., 2009), predicted trends and topic modeling (Gupta et al., 2022), supervised topic models (Li et al., 2023; Wang et al., 2021; Mcauliffe and Blei, 2007), latent Dirichlet co-clustering (Shafiei and Milios, 2006), and LDA-based work in bioinformatics (Gurcan and Cagiltay, 2022; Garg and Rangra, 2022; Kawai et al., 2022).

LDA's versatility also becomes evident across diverse domains, finding applications in various areas. For instance, it contributes to uncovering roles within Social Network Analysis (Kushwaha et al., 2022) through models like the Author-Recipient-Topic (ART) model (McCalum et al., 2007). In the context of emotion topic modeling, techniques such as Pairwise-Link-LDA are employed (Amouzgar et al., 2022; Nallapati et al., 2008). Moreover, LDA plays a significant role in tasks like automatic essay grading (He et al., 2022; del Gobbo et al., 2023), text categorization (Eligüz et al., 2022; Edara et al., 2023), and information retrieval (Hamarde and Proenca, 2023; Deveaud et al., 2014; Wei and Croft, 2006). Its utility extends even to enhancing cybersecurity measures (Alagheband et al., 2020; Al Moubayed et al., 2017; Sufi, 2023), including anti-phishing initiatives (Rao and Pais, 2019) and

spam detection (Maurya et al., 2023). A concrete demonstration of LDA's effectiveness can be observed through a practical application on real-world data (Blei et al., 2003).

Furthermore, recent years have seen a notable rise in the prominence of neural topic models within the field of topic modeling. The integration of deep neural networks (Emmert-Streib et al., 2020) into existing techniques has led to significant successes, as evidenced by Zhao et al. (2021), Cao et al. (2015) and Larochelle and Lauly (2012). Additionally, the incorporation of word embeddings into traditional models, like LDA, has highlighted the potential of harnessing these representations (Liu et al., 2015; Nguyen et al., 2015; Qiang et al., 2017; Shi et al., 2017).

It is interesting to note that there has been a recent upsurge in the development of topic modeling techniques centered around embedding models. This trend is exemplified in various studies (Bianchi et al., 2020b; Dieng et al., 2020). Notably, the Correlated Topic Model (CTM) emphasizes the advantage of relying on pre-trained language models, suggesting the potential for enhanced topic modeling (Bianchi et al., 2020a).

To simplify the topic construction process, recent approaches have used a clustering strategy based on word and document embeddings, e.g., Sia et al. (2020) and Angelov (2020). This strategy decouples the generation of clusters from the creation of topic representations, providing flexibility in the topic modeling process by building upon clustering embeddings. BERTopic (Grootendorst, 2022) enhances this capability by introducing a class-based variant of TF-IDF for topic representations. BERTopic operates on the premise that documents sharing the same topic exhibit a semantic similarity. To achieve this, it employs the Sentence-BERT (SBERT) framework (Reimers and Gurevych, 2019) for document embeddings, enabling the creation of vector-based representations that facilitate semantic comparisons.

3. Research questions

Our investigation relies on a distinctive dataset we meticulously compiled, comprising thousands of abstracts related to sustainable energy research. To organize and classify this extensive dataset into coherent topics, we contrast the effectiveness of a Latent Dirichlet Allocation (LDA) algorithm with BERTopic. Our key aim is to introduce a mathematical optimization method that facilitates the identification of the optimal number of topics. Ultimately, this approach will enable a more profound comprehension of the content presented in articles on sustainable energy research, offering insights into their evolution over time and uncovering enduring prominent themes. The specific research questions addressed in this paper are as follows:

1. Are traditional topic modeling methods less effective than modern approaches? We study this by comparing Latent Dirichlet Allocation (LDA) with BERTopic.
2. How do the distributions of semantic similarity scores change with the number of topics?
3. What is the optimal number of topics for LDA? This is investigated for different utility scores based on perplexity and coherence measures.
4. What are the core themes and subjects studied in sustainable energy research? Are there changes of topics over time?

4. Methods and data

This section provides an overview of the data used in our study, outlines the pre-processing methods employed as depicted in Fig. 1, and introduces the chosen approach for conducting topic modeling.

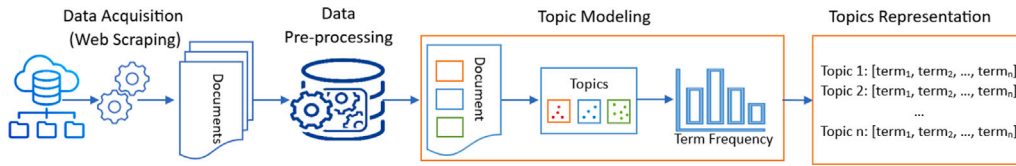


Fig. 1. The procedure for performing topic modeling on sustainable energy data entails retrieving information from reputable databases covering the period from 1975 to 2023, encompassing abstracts from 7016 journals. The initial steps involve pre-processing, which includes tasks like removing metadata, normalizing text, eliminating punctuation, tokenizing, removing stopwords, and lemmatizing. Following this, the refined dataset is fed into a topic modeling algorithm like LDA or BERTopic, with a specific emphasis on fine-tuning parameters. The concluding phase involves evaluating the model's output (topics) using various metrics to derive meaningful conclusions.

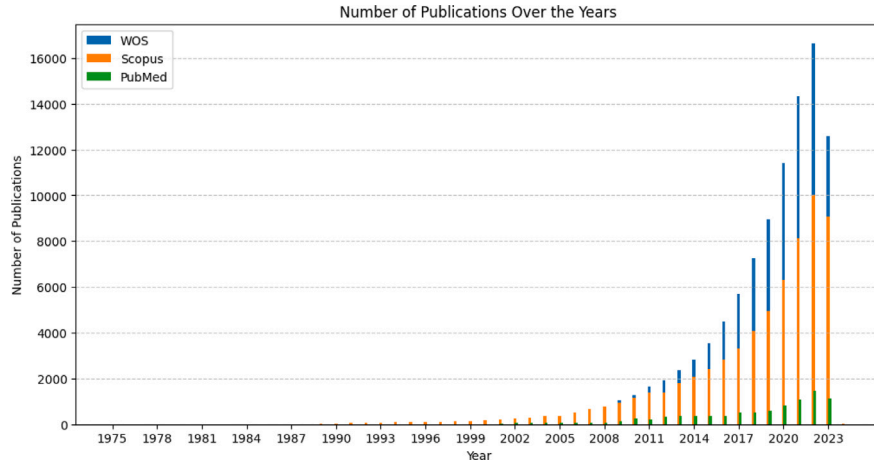


Fig. 2. The growth in the number of sustainable energy publications over the years for PubMed, Scopus, and Web of Science.

Table 1

A summary of our text data comprising 173,146 unique articles (abstracts) from 7016 journals.

Dataset	Retrieved articles	Unique articles	Number of journals
PubMed	9999	9862	543
Scopus	69,647	64,386	5396
Web of science	104,010	98,898	1077
Total	183,656	173,146	7016

4.1. Data acquisition

For our analysis, we extract text data corresponding to scientific abstracts about sustainable energy research from three databases: PubMed (PubMed, 2013), Scopus (Scopus, 2013), and Web of Science (WOS) (WoS, 2013). The publication history of these articles is shown in Fig. 2. Overall, the articles span from 1975 to 2023 and are drawn from 7016 journals. This figure also highlights the significance of sustainable energy research through the increasing number of its publications each year. After eliminating duplicates and non-English articles, we end up with 9862 unique articles from PubMed, 64,386 from Scopus, and 98,898 from Web of Science as shown in Table 1.

In total, the articles extracted from these databases amounted to a collection of 173,146 unique articles. The irrelevant metadata such as author names, language, document type, times cited, publisher-related information, journal abbreviation, volume, issue, categories, etc., have been excluded from the articles. We subsequently consolidated pertinent metadata, including article title, keywords, abstract, publication date, and source title, which hold insights and implications for this study, for further pre-processing and analysis.

4.2. Pre-processing

In the domain of natural language processing, and text analysis, pre-processing plays a pivotal role in refining raw text data for

subsequent analysis and modeling. Initially, text is converted to lowercase, a seemingly minor adjustment that harmonizes the text and ensures consistent treatment of words, irrespective of their original casing. Next, the process moves to punctuation removal, where all non-alphanumeric characters like commas, periods, and parentheses are stripped away, thereby sharpening the focus on meaningful words, which might otherwise introduce noise into the analysis. The next step involves tokenization, breaking down the text into individual tokens or words, enabling a more granular level of analysis.

Further, we incorporated the analysis of bi-grams and tri-grams, fine-tuning parameters such as “min_count” and “threshold” to regulate the strictness of word combinations, with higher values introducing greater complexity to the amalgamation of words. Bi-grams refer to pairs of words that exhibit frequent co-occurrence within a document, while tri-grams are composed of three words that often occur together.

Subsequently, the focus shifts to stopwords removal, where common words such as “the” and “and” are eliminated. Additionally, words with a length of less than 3 letters are excluded, as this step helps eliminate many abbreviations found in our dataset with 2-letter lengths, further refining the text for analysis. Lastly, lemmatization comes into play, aiding in feature reduction and reducing words to their base or root form, ensuring that variations of the same word, such as “running” and “run”, are treated as identical entities. Collectively, these pre-processing steps elevate the quality and coherence of text data, making it more amenable to shaping raw text into a manageable and informative format for various natural language processing tasks.

4.3. Topic modeling approaches

In this study, we utilize two topic models for our analysis: LDA and BERTopic. The number of latent topics denoted as K , is a free parameter in both models. The remaining parameters for LDA, specifically β , which represents the topic multinomial distribution for topic k , and α , representing the topic mixture proportions for document d , inferred from the data. It is important to note that the two models differ in the methodology used to infer these latent parameters.

- Latent Dirichlet Allocation (LDA): is a generative model for documents in which the mixture proportions θ_d are regarded as random variables drawn from a Dirichlet prior distribution (Blei et al., 2003).

LDA models each of the D documents as a mixture over K latent topics, describing multinomial distributions across a W word vocabulary. The generative process for basic LDA, as described by Porteous et al. (2008), involves selecting a topic $z_{ij} \sim \text{Mult}(\theta_j)$, for each of the N_j words in document j and choosing a word $x_{ij} \sim \text{Mult}(\phi_{z_{ij}})$.

Nonetheless, the operational definition of LDA can also be articulated as depicted in Eq. (1).

$$P(W, Z, \theta, \phi; \alpha, \beta) = \prod_{j=1}^M P(\theta_j; \alpha) \prod_{i=1}^K P(\phi_i; \beta) \prod_{i=1}^N P(Z_j, t | \theta_j) \times P(W_j, t | \phi_j, t) \quad (1)$$

where the term $P(W, Z, \theta, \phi; \alpha, \beta)$ represents the probability that a specific document originates from a hypothetical LDA (Latent Dirichlet Allocation) machine. These configurations are pivotal in unveiling the latent topics within the document. α and β are associated with Dirichlet distributions, whereas θ and ϕ are related to Multinomial distributions. In this context, ‘Z’ signifies the list of topics, while ‘W’ denotes the collection of words. The value of ‘P’ is determined by four independent probabilities, as detailed on the right-hand side of Eq. (1). Importantly, ‘K’ is one of the three model hyperparameters, representing the number of topics, while ‘M’ and ‘N’ respectively indicate the number of documents and the word count within a given document.

In the process of modeling our documents, we utilize a specific vocabulary, which we define as follows. We create a *corpus* that encompasses all the documents under analysis, with D , representing the total number of documents. Here, a *document* corresponds to an individual abstract sourced from the compiled dataset.

Each document, denoted as d comprises a collection of W words, and the aggregation of distinct words found across all documents, spanning various journals papers, constitutes the model’s dictionary or vocabulary, referred to as V . Additionally, each document is linked to a journal and its publication year, introducing further layers of document categorization.

- BERTopic: BERTopic utilizes a modified TF-IDF representation to capture the significance of terms within specific topics rather than solely at the document level. The conventional TF-IDF method combines two significant statistics: term frequency and inverse document frequency (Joachims et al., 1997):

$$W_{t,d} = t f_{t,d} \cdot \log \left(\frac{N}{d f_t} \right) \quad (2)$$

In this equation, term frequency represents the frequency of term t in document d , while the inverse document frequency measures the information that a term provides to a document. It is calculated by taking the logarithm of the total number of documents in a corpus N divided by the number of documents containing term t .

This procedure has been extended to clusters of documents in BERTopic. Initially, it treats all documents in a cluster as a single document by concatenating them. Then, it adjusts the TF-IDF to accommodate this representation by translating documents into clusters (Grootendorst, 2022):

$$W_{t,c} = t f_{t,c} \cdot \log \left(1 + \frac{A}{t f_t} \right) \quad (3)$$

In this equation, term frequency models the frequency of term t in a class c , which essentially combines the collection of documents into a single document for each cluster. The inverse document frequency is replaced by the inverse class frequency to assess how

much information a term contributes to a class. It is calculated by taking the logarithm of the average number of words per class A divided by the frequency of term t across all classes. To ensure positive values, one is added to the division within the logarithm. This class-based TF-IDF method models the importance of words within clusters rather than individual documents, enabling the generation of topic-word distributions for each cluster of documents. Ultimately, by iteratively merging the class-based TF-IDF representations of the least common topic with its most similar one, it can reduce the number of topics to a user-specified value.

LDA operates as a probabilistic generative model, assuming that each document comprises a blend of distinct topics, with each word in the document assigned to one of these topics. The model establishes the probability of each word given a topic and the probability of each topic given a document. It demonstrates scalability, capable of handling large corpora, although training time increases with the size of the corpus and the number of topics. LDA effectively manages varying document lengths due to its probabilistic nature, allowing flexibility in document structures. However, it is sensitive to hyperparameters, such as the number of topics and Dirichlet priors, making finding optimal values challenging. LDA naturally accommodates multiple topics per document, reflecting real-world scenarios.

On other hand, BERTopic leverages transformer-based models to generate context-aware embeddings for text documents. It utilizes clustering algorithms on these embeddings to identify topics, providing more contextually rich representations. While BERTopic may offer somewhat less direct interpretability compared to LDA due to its reliance on complex transformer-based models, it compensates with contextually rich topics and meaningful representations. BERTopic can be computationally intensive, especially for large datasets, due to the use of transformer-based embeddings. Similar to LDA, BERTopic is sensitive to hyperparameters, particularly those related to clustering algorithms used in conjunction with transformer embeddings. It adeptly handles multiple topics per document, providing a nuanced view of topic distributions. BERTopic proves valuable in applications requiring clustering and a deeper understanding of the contextual representation of topics within a corpus.

4.4. Fine-tuning of the LDA model

An essential characteristic shared by topic models like LDA and BERTopic is their capacity to autonomously determine the number of topics during modeling or to require a specified number in the training phase. However, identifying the optimal number of topics remains a significant challenge without a straightforward solution. As highlighted in the previous section, this large number may hinder humans from focusing on all these many topics.

In our effort to tackle this challenge, we endeavored to limit the number of topics during the training of the LDA model to a specific manageable quantity. Various scores were employed to assess the generative topic representations of each model, aiming to identify the most coherent representation, as detailed in the next section.

In this study, we conducted a random search to determine the optimal values for the five hyperparameters essential for Latent Dirichlet Allocation (LDA): the number of topics (K), alpha, beta, passes, and chunk size, as outlined in Table 2. The hyperparameter of a number of topics informs the model about the desired number of topics to discover. Adjusting this can have a significant impact on both coherence and perplexity. Choosing an appropriate number of topics is crucial for obtaining meaningful and coherent topics. Alpha controls the document-topic density. Adjusting alpha can influence topic sparsity, and finding the right balance is important for achieving coherent topics. Higher alpha values result in more diverse topics, while higher beta values lead to topics comprising more words. The number of passes determines the passes through the entire corpus during training.

Table 2
Parameter configuration for LDA Models.

Parameter	Values
# of topics	4, 8, 12, 16, 20, 24, 28, 32
Alpha	0.01, 0.1, 0.3, 1
Beta	0.01, 0.1, 0.3, 1
Passes	10
Chunk size	100, 500, 1000

Increasing the number of passes may improve the quality of topics by allowing the model to refine the topic assignments. While the chunk size controls the number of documents processed in each training chunk. It affects the speed and memory requirements during training. Adjusting this might impact the convergence speed and, indirectly, coherence and perplexity. The search for the optimal number of topics initiated with a range from 4 to 32, incrementing by 4. During the initial learning step, K was pre-defined, and the search for beta, alpha, passes, and chunk size was subsequently executed. Furthermore, each model's data underwent two encoding processes: one using the Bag-of-Words (BoW) representation and another using Term-Frequency Inverse Document Frequency (TF-IDF). BoW represents documents as vectors, focusing on word counts and ignoring word order. TF-IDF considers word importance in a document relative to their occurrence in the entire corpus. This resulted in a total of 150 experiments for each encoding representation, totaling 300 experiments. To enhance the interpretability of information extracted from a fitted LDA topic model, we employed pyLDAvis to generate an intertopic distance map (Islam, 2019).

4.5. Evaluation scores

Despite the prevalent belief that topic modeling uncovers meaningful and valuable latent concepts, the validation of this assumption is challenging due to the unsupervised nature of topic discovery. This lack of a definitive set of topics for every corpus necessitates the incorporation of external scores for evaluating the identified topics in latent spaces.

In general, the evaluation of models can be done via two approaches: extrinsic evaluation and intrinsic evaluation. Extrinsic evaluation requires testing the models on real tasks and analyzing the final accuracy, which is the best way to understand how different models affect the task. However, it can be slow and computationally expensive. On the other hand, intrinsic evaluation involves finding a metric to assess the model performance based on its internal characteristics, without considering the specific task it will be used for.

For our analysis, we employ different evaluation scores, such as perplexity, and coherence, to pinpoint the most suitable number of topics for the fine-tuned LDA model. Later, we introduce a mathematical optimization to determine the optimal number of topics for the LDA model, which we will also utilize for BERTopic, which is based on those scores. In the following, we discuss the four scores we are using for our analysis.

4.5.1. Perplexity

Perplexity (PP) serves as an intrinsic evaluation score that is used to evaluate how well a model predicts a sequence of words. A good model should have low perplexity and assign high probabilities to real and syntactically correct sentences while assigning high perplexity and low probabilities to fake, incorrect, or infrequent sentences. Perplexity also serves as an indicator of the topic model's predictive performance on novel or unseen data. It mirrors the model's ability to generalize. A lower perplexity score indicates the model's confidence and accuracy in predictions, while a higher perplexity score suggests uncertainty and inaccuracy in its predictions.

Although the dictionary defines perplexity as "the state of being puzzled or confused", in AI, it refers to a statistical measure that

assesses the ability of a language model to predict the level of confusion in a given text sample.

In general, perplexity is fast to calculate, robust to outliers, and can help estimate a language model's uncertainty, which makes it a useful tool for comparing different models, identifying issues in datasets, and fine-tuning model parameters. However, it is not an accurate measure for the final evaluation and may be challenging to compare across datasets. Additionally, it might favor models trained on outdated datasets and is not always precise in measuring a model's performance. Therefore, it is important to use perplexity as an auxiliary measure and evaluate models using other metrics as well. Perplexity (PP) was defined by Klakow and Peters (2002) as:

$$PP = \exp \left(\frac{1}{N} \sum_{i=1}^N \log p(w_i | h_i) \right) \quad (4)$$

Here, w_i represents the word at the i th position in the test set, and h_i denotes the preceding history.

4.5.2. Coherence

In general, a coherence score quantifies the degree of semantic interconnection among words within a given topic. It gauges the extent to which the words within a topic are meaningfully related, aligning with our human sense of a well-structured topic. In other words, it evaluates the interrelation among the top-k words within a given topic. A higher coherence score indicates that the topic is cohesive, clear, and pertinent, while a lower score suggests the topic lacks clarity, contains noise, or is irrelevant.

In this study, we use various coherence scores, including UCI Coherence, UMass Coherence, and CV Coherence (Bouma, 2009; Lau et al., 2014; Röder et al., 2015; Webber et al., 2010; Syed and Spruit, 2017), to compare word distributions within the generated topics against the reference corpus. The evaluated metrics for topic coherence involve selecting the top N words within a given topic and then summing a confirmation measure for every pair of words. The confirmation measure is dependent on a single pair of top words.

1. UCI Coherence: This score assesses the coherence of topics through an evaluation of pairwise word similarity within each topic. Elevated values signify more cohesive topics. The UCI score relies on a sliding window and utilizes pointwise mutual information (PMI) for all word pairs among the specified top words.

$$C_{UCI} = \frac{2}{N \cdot (N - 1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \text{PMI}(w_i, w_j) \quad (5)$$

$$\text{PMI}(w_i, w_j) = \log \frac{P(w_i, w_j) + \epsilon}{P(w_i) \cdot P(w_j)} \quad (6)$$

Here $P(w)$ represents the probability of encountering the word w within the sliding window, while $P(w_i, w_j)$ denotes the probability of both words w_i and w_j appearing together in the sliding window. That means probabilities are estimated by analyzing the frequency of word co-occurrences. These frequencies are generated from documents formed through a sliding window that scans the dataset in use, serving as an external reference corpus.

2. UMass Coherence: The UMass coherence score utilizes a different algorithm than UCI coherence. Generally, lower (more negative) values in the UMass coherence score suggest more coherent topics. Consequently, a more negative UMass coherence score indicates more clearly defined and interpretable topics. The UMass score is based on document co-occurrence counts, a one-preceding segmentation, and a logarithmic conditional probability as a confirmation measure.

$$C_{UMass} = \frac{2}{N \cdot (N - 1)} \sum_{i=2}^N \sum_{j=1}^{i-1} \log \frac{P(w_i, w_j) + \epsilon}{P(w_j)} \quad (7)$$

Here, $P(w_i, w_j)$ represents the frequency of co-occurrence of words w_i and w_j in documents, while $P(w_i)$ denotes the frequency of occurrence of the word w_i in isolation. That means probabilities for words are calculated by considering the document frequencies of the original documents utilized in the topic-learning process. ϵ is introduced to prevent the logarithm of zero. Both UCI and UMass can be applied using either the arithmetic mean or solely the sum of individual elements. This is because the mean does not impact the ranking of evaluated topics that share the same number of top words.

3. CV Coherence: The context vector (CV) coherence score is yet another coherence score. A higher scores are considered relatively favorable, suggesting a moderate level of coherence within the topics. Also the CV score relies on a sliding window, one-set segmentation of the top words, and an indirect confirmation measure employing normalized pointwise mutual information (NPMI) and cosine similarity.

$$\vec{v}(W') = \left\{ \sum_{w_i \in W'} \text{NPMI}(w_i, w_j)^\gamma \right\}_{j=1, \dots, |W|} \quad (8)$$

$$v_{ij} = \text{NPMI}(w_i, w_j)^\gamma = \left(\frac{\log \left(\frac{P(w_i, w_j) + \epsilon}{P(w_i) \cdot P(w_j)} \right)}{-\log(P(w_i, w_j) + \epsilon)} \right)^\gamma \quad (9)$$

A context vector for a word, w , is constructed through the analysis of word co-occurrence frequencies within context windows. These windows encompass all words situated ± 5 tokens around instances of the word w . The optimal correlation with human topic coherence ratings is achieved by representing the elements of these vectors as normalized Pointwise Mutual Information (NPMI).

Furthermore, superior results can be obtained by constraining word co-occurrences to those within the same topic, forming what is referred to as the top word space. Consequently, the j th element of the context vector (v_j) associated with the word w_i is determined by NPMI, with an augmented emphasis on higher NPMI values through an increase in the parameter γ (Röder et al., 2015). The confirmation measures between pairs of these context vectors are assessed through vector similarity metrics such as cosine similarity, Dice coefficient, or Jaccard index. These similarity scores are averaged across all pairs of top words within a given topic.

It is important to emphasize that categorizing perplexity or coherence scores as either good or bad lacks a universal standard. The assessment of coherence scores and their perceived quality is contingent upon the specific dataset from which they are derived. For instance, a coherence score of 0.5 may be considered satisfactory in one context but deemed unacceptable in another.

5. Results

In this section, we present our results. First, we study topic modeling for an unconstrained selection of topics. Then we study different evaluation scores and present an optimization method for the identification of the optimal number of topics. Finally, we study the evolution of topics over time.

5.1. Unconstraint selection of topics

The first analysis we perform is a topic modeling that does not put constraints on the number of topics for the selection. Specifically, we use preprocessed data as input for both LDA and BERTopic and conduct an analysis giving the autonomy to automatically determine the respective number of topics. As a result, LDA generates 100 topics, while

BERTopic finds 1101 topics. The results of this analysis are visualized in Figs. 3(a) and 4(a) respectively. The intertopic distance maps illustrate the connections or disparities among various topics within a topic model. These visual representations depict the degree of distinctness or correlation between topics. The distances between points in these maps symbolize the similarity or dissimilarity between the respective topics. Figs. 3(b) and 4(b) show the top four terms associated with the first 8 topics for both LDA (left) and BERTopic (right). From the bar charts, one can see the weights of those terms. Here, we displayed only the first eight topics with their top four associated terms for LDA out of a total of 100, because a visualization of all 1101 topics is incomprehensible.

Finally, in Figs. 3(c) and 4(c) we showcase the similarity among the generative topics for LDA and BERTopic. The heatmap shows the cosine similarity, representing the extent to which word distributions of distinct topics align. This visual representation offers a rapid assessment of the degree of overlap or disparity between the word distributions of topics.

The proceeding analysis allows us to shed light on a fundamental challenge in topic modeling. In particular, both models can yield a large number of generated topics, which presents a considerable hurdle in their effective evaluation. This challenge is further exacerbated when either there is a significant overlap between the generated topics or when the objective is to focus on a specific set of topics. This calls for nuanced evaluation scores and strategies to address the complexity introduced by the potential abundance and overlap of generated topics.

In the following, we study four different evaluation scores upon which we will build a mathematical optimization that allows us to obtain the optimal number of topics for topic modeling.

5.2. Explorative analysis of evaluation scores

In this section, we present results for four evaluation scores: Perplexity (PP), UCI Coherence, UMass Coherence, and CV Coherence. These metrics, introduced in Section 4.5, allows us to summarize information represented by different topics.

In Figs. 5 and 6 the scores Perplexity (PP), UCI Coherence, UMass Coherence and CV Coherence are shown for 150 different experiments based on different parameter settings (see Section 4.4 for details). It is important to note that from these figures, one can see that each score recommends a different number of topics. For instance, Fig. 5 indicates that the optimal number of topics for Perplexity (PP) is 4 with $\alpha = 0.3$, $\beta = 0.01$, and chunk size = 100. On the other hand, UCI coherence, shown in Fig. 6(a), suggests that the ideal number of topics is 32 with $\alpha = 1$, $\beta = 0.3$, and chunk size = 1000; UMass coherence in Fig. 6(b) suggests 32 topics with $\alpha = 0.01$, $\beta = 1$, and chunk size = 100; and CV coherence in Fig. 6(c) recommends 20 topics with $\alpha = 0.01$, $\beta = 1$, and chunk size = 100.

In order to emphasize the distributional character of the scores Perplexity (PP), UCI Coherence, UMass Coherence, and CV Coherence, we show in Fig. 7 their underlying distribution in dependence on the number of topics. As one can see, there are large variations of the distributions within and between the four scores. For instance, while the median value of PP remains largely unchanged for UMAS it is decreasing with larger topic numbers. Yet a different behavior can be observed for UCI and CV which both exhibit an inverted “U” shape. Furthermore, also the spread of the distributions is topic- and score-dependent. While for PP the largest interquartile range (IQR) is observed for 16 topics for UCI, UMAS, and CV it is 24, 24, and 20 respectively. Overall, Fig. 7 emphasizes that each score is a random variable which is part of the difficulty in determining the optimal number of topics for topic modeling.

To gain an understanding of the overlap of the four scores and some hyperparameters, we show in Fig. 8 their correlations. Regarding the scores, we see that UCI and UMass have the highest correlation whereas UCI and CV have the lowest. This is interesting because PP is a different score than the three coherences sores but CV is quite different

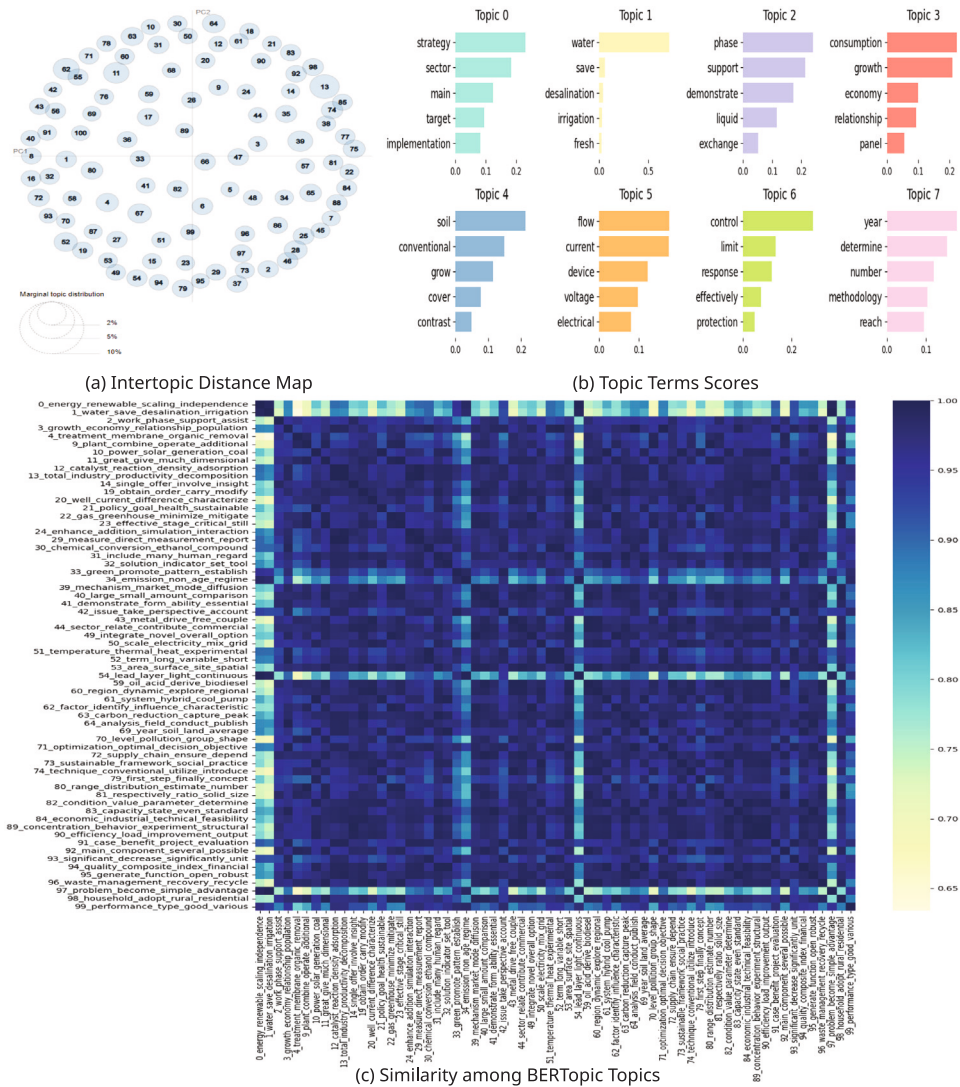


Fig. 3. Topic modeling using LDA with topics dynamically chosen by the model. (a) All topics of the LDA model. (b) Top four terms for the first 8 topics. (c) Correlation similarity between topics.

to UCI and UMass. For the hyperparameters, we observe in general very small correlations which underscores their different character. For the mixed correlation between scores and hyperparameters, we find a higher correlation for PP and beta, UCI and alpha, and CV and beta, and a strong anti-correlation for UMass and num_topic.

Finally, in order to summarize the above results from the 150 individual experiments, we show in Table 3 the highest and lowest values for each score across the studied number of topics. Furthermore, in Fig. 9, we show a visual overview depicting the top scores across the examined range of topics. Specifically, in Fig. 9(a) an upward trend in coherence scores with an increase in the number of topics is shown. This increase diminishes as the number of topics becomes higher. However, the topics and terms generated by LDA over the TF-IDF-encoded data (see Fig. 9(b)) in our analysis were found to be unsatisfactory, involving substantial convergence among topics, encompassing predominantly uncommon and infrequently used terms. This representation gives importance to words in a document relative to their occurrence across the entire corpus, predominantly favoring rare words, bi-grams, and tri-grams. As a result, Fig. 9(b) exhibits irregular patterns. Consequently, in the subsequent sections, we transition away from this representation and instead, we opt for the Bag-of-Words (BoW) representation.

5.3. Determining the optimal number of topics

Despite the various coherence measures proposed in the literature as methods for ranking topics and exploring the space of topic coherence, determining the optimal number of topics remains a challenge due to various reasons. One is that each score may suggest a different optimal number of topics. As an example, this lack of harmony between coherence scores but also perplexity can be seen in Table 3 and Fig. 9(a). Specifically, if one would use the four scores individually as optimization criteria, for our data, the optimal number of topics would be 4 (perplexity), 32 (UMass coherence), 20 (CV coherence) and 32 (UCI coherence). Another complication is that different coherence scores and perplexity should not even result in the same result due to their different perspective they provide on the data. To address this issue, we formulate the determination of the number of topics as a mathematical optimization problem. A summary of this is outlined in Algorithm 1 and in the following, we provide a detailed discussion.

From Figs. 5–7 Table 3 one can see that this is not straight forward to combine these scores due a number of issues. First, the different scores assume values in considerably different numerical ranges. For instance, for 4 topics in Table 3 the highest value of PP is -33.78 while the lowest is -7.461 . In contrast, the corresponding values for UCI are 0.193 and 0.128. Second, some of the scores only assume positive



Fig. 4. Topic modeling using BERTopic with topics dynamically chosen by the model. (a) Corresponds to the BERTopic model, showcasing its topics. (b) Top four terms for the first 8 topics. (c) Explores the correlation similarity between topics.

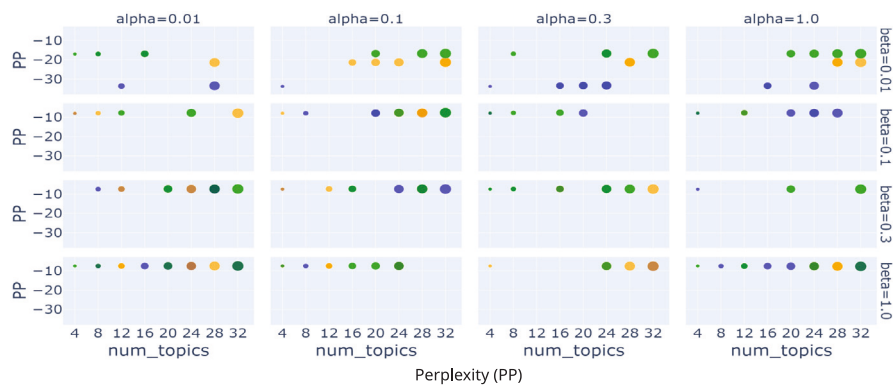


Fig. 5. Perplexity scores for LDA across various experiments with different hyperparameter settings utilizing Bag of Words (BoW) encoded data. The colors indicate different chunk sizes: blue = 100, yellow = 500, green = 1000.

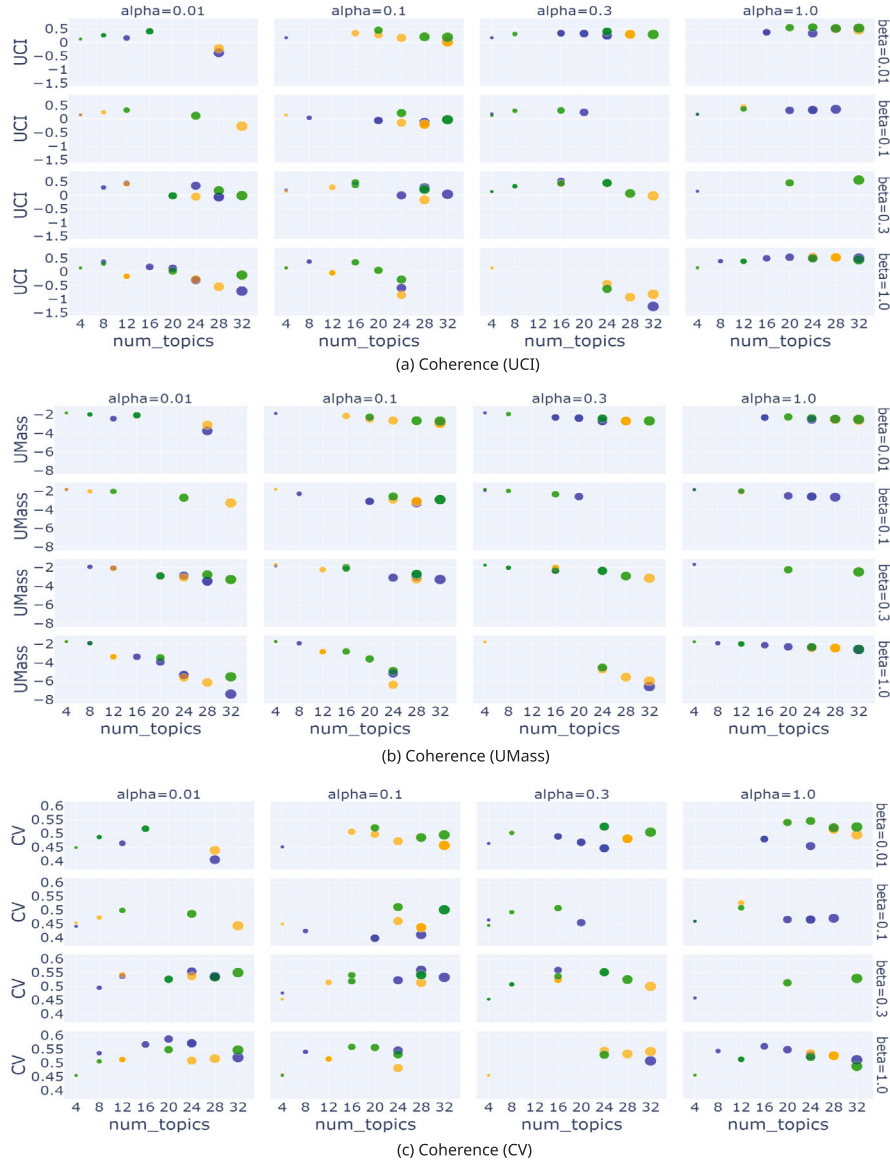


Fig. 6. Coherence scores for LDA across various experiments with different hyperparameter settings utilizing Bag of Words (BoW) encoded data. The colors indicate different chunk sizes: blue = 100, yellow = 500, green = 1000.

values, such as CV, while others assume only negative values, such as PP or UMass. Yet differently, UCI assumes mixed values. Third, some scores are considered optimal for maximum values, such as UCI, UMass, or CV, while PP is optimal for its minimum value.

This discussion indicates the necessity of normalizing the scores before incorporating them into the formulation of an optimization problem, as utilizing them in their raw form is prone to numerical issues and potential failure. For this reason, we, first, adjust for the different meanings of optimality and, second, we conduct a numerical normalization of the scores.

For obtaining a coherent meaning of optimality among perplexity and the coherence scores, we need to adjust the perplexity (PP) scores as follows

$$PP' = (-1)PP. \quad (10)$$

As a result of this operation all scores, i.e., PP', UCI, UMass, and CV are optimal for the maximum values of the corresponding scores.

Next, we introduce further definitions we need for the normalization of the scores. To simplify the notation, we call all scores by x_i with the understanding that this corresponds to one of the four different

Algorithm 1 Determining the Optimal Number of Topics

- 1: Define the set of all different score values (S) for 150 different experimental settings:
- 2: $S = \{x_i \in \mathbb{S} \mid 1 \leq i \leq 150\}$
- 3: **Normalize Scores:**
- 4: **For** each score $x_i \in S$:
- 5: Average Normalization (norm 1):
- 6: $S_i = \frac{x_i - \mu}{\sum_{j=1}^n x_j}$
- 7: Min-Max Normalization (norm 2):
- 8: $S_i = \frac{x_i - \min(x)}{\max(x) - \min(x)}$
- 9: T-Score Normalization (norm 3):
- 10: $S_i = \frac{x_i - \mu}{\sigma}$
- 11: Calculate the Combined Score (CS) for each experiment:
- 12: **For** each experiment i :
- 13: $CS_i = \sum_{j=1}^n \lambda S_{ij}$
- 14: **Determine the Best Combined Score:**
- 15: Best CS = $\arg \max(CS)$
- 16: **Determine the Optimal Number of Topics (OP):**
- 17: OP = $\arg \max_i CS_i$

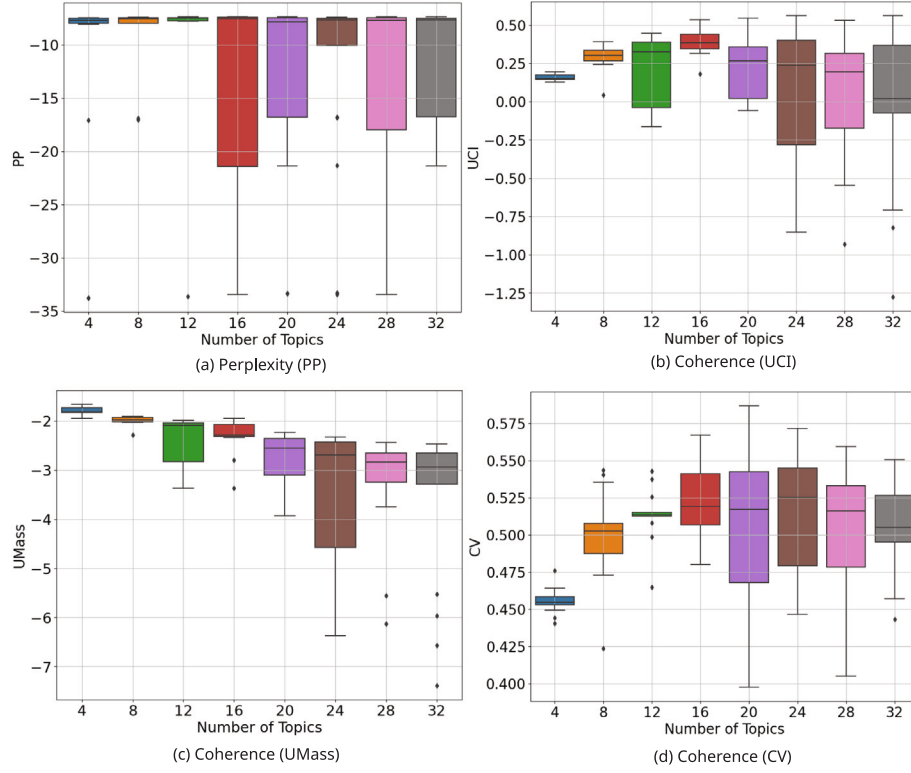


Fig. 7. Box plots for LDA showing Perplexity (PP), UCI, UMass, and CV Coherence scores across various numbers of topics. For this analysis, we used a Bag of Words (BoW) representation of the data.

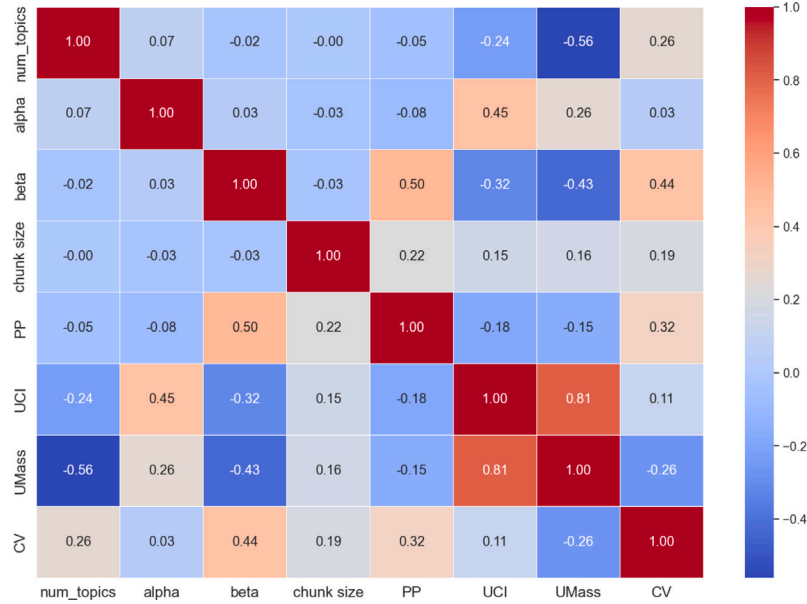


Fig. 8. Interrelations between hyperparameters and the corresponding perplexity and coherence scores for LDA (Bag of Words (BoW) representation of encoded data).

scores PP', UCI, UMass, and CV. The set of all different score values for 150 different settings (see Section 4.4 describing the conducted experiments) is given by

$$S = \{x_i \in \mathbb{S} \mid 1 \leq i \leq 150\} \quad (11)$$

The scores are normalized using the following techniques:

1. *Average Normalization (norm 1):*

$$S_i = \frac{x_i - \mu}{\sum_{j=1}^n x_j} \quad (12)$$

2. *Min-Max Normalization (norm 2):*

$$S_i = \frac{x_i - \min(x)}{\max(x) - \min(x)} \quad (13)$$

3. *T-Score Normalization (norm 3) (Emmert-Streib and Dehmer, 2019):*

$$S_i = \frac{x_i - \hat{\mu}}{\hat{\sigma}} \quad (14)$$

Table 3

The highest and lowest scores for each metric across varying numbers of topics based on the configuration outlined in Section 4.4.

Perplexity and coherence scores for topics 4,8,12,16,20,24,28,32 based on BoW encoding								
Topics size	Perplexity		UCI coherence		UMass coherence		CV coherence	
	Highest	Lowest	Highest	Lowest	Highest	Lowest	Highest	Lowest
4	-33.780	-7.461	0.193	0.128	-1.951	-1.664	0.476	0.441
8	-17.011	-7.403	0.390	0.042	-2.288	-1.909	0.544	0.424
12	-33.634	-7.372	0.446	-0.164	-3.377	-1.995	0.543	0.465
16	-33.477	-7.339	0.533	0.181	-3.372	-1.952	0.567	0.480
20	-33.381	-7.364	0.546	-0.057	-3.929	-2.234	0.587	0.398
24	-33.463	-7.373	0.561	-0.851	-6.370	-2.333	0.572	0.447
28	-33.480	-7.336	0.531	-0.931	-6.132	-2.440	0.559	0.405
32	-21.351	-7.365	0.562	-1.274	-7.380	-2.473	0.550	0.443
Perplexity and coherence scores for topics 4,8,12,16,20,24,28,32 based on TF-IDF encoding								
Topics size	Perplexity		UCI coherence		UMass coherence		CV coherence	
	Highest	Lowest	Highest	Lowest	Highest	Lowest	Highest	Lowest
4	-50.435	-8.909	1.012	-0.779	-7.757	-1.898	0.589	0.429
8	-54.537	-9.042	1.915	-1.153	-9.380	-2.072	0.783	0.310
12	-50.516	-9.100	2.690	-2.213	-13.292	-2.242	0.643	0.225
16	-57.228	-9.174	1.526	-2.639	-15.169	-2.294	0.517	0.189
20	-51.885	-9.392	0.137	-3.777	-15.662	-2.228	0.447	0.149
24	-52.071	-9.272	0.504	-3.919	-15.597	-2.479	0.518	0.173
28	-59.319	-9.314	0.473	-3.133	-16.019	-2.135	0.499	0.186
32	-52.234	-9.317	0.276	-3.198	-16.121	-2.055	0.450	0.165

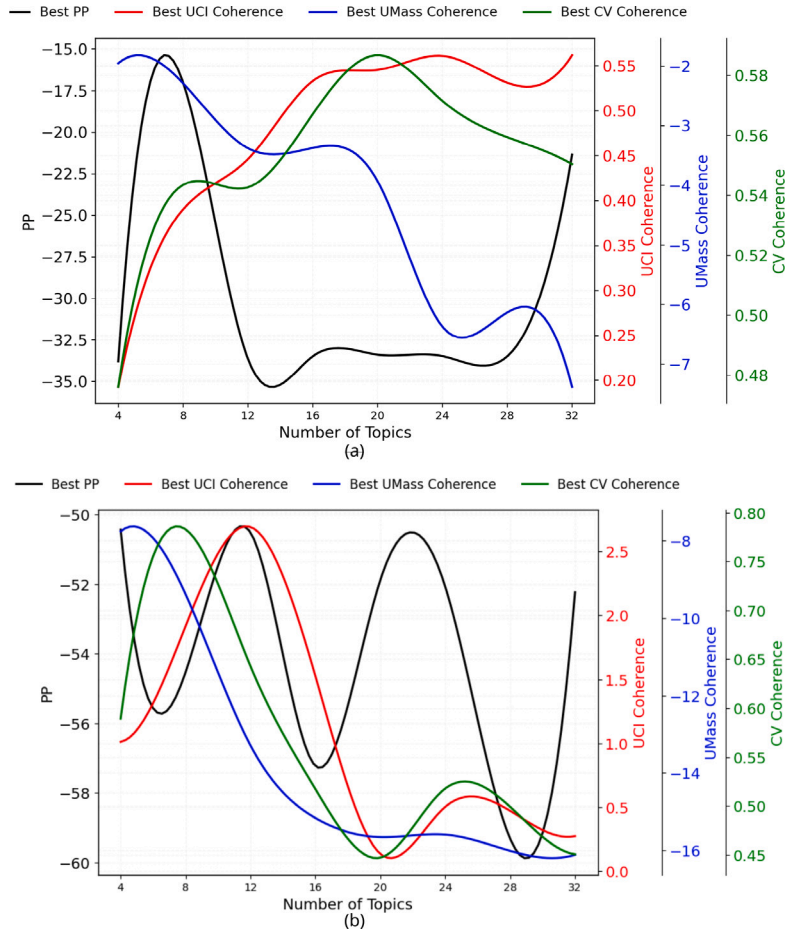


Fig. 9. Scores for PP, UCI, UMass and CV across different numbers of topics (see Section 4.4). Top: Shown are scores for BoW data encoding. Bottom: Scores for data encoded using TF-IDF.

From S_i , we determine the combined score (CS) for each experiment using the following formulation:

$$CS_i = \sum_{j=1}^n \lambda S_i \quad (15)$$

Here, the parameter λ serves as a regularizer for the normalized score. However, assigning different values to λ for each score can significantly influence exploring diverse topics and uncover more intriguing observations within the recommended number of topics, as evidenced in Table 4. The λ coefficients $[v_1, v_2, v_3, v_4]$ serve as weighting factors,

ranging from 0 to 1, applied to the normalized scores of perplexity, UCI, CV, and UMass coherence, respectively. Negative values of v_1 and v_4 indicate optimal values for perplexity and UMass coherence, respectively (with lower values being preferable, as outlined in Sections 4.5.1 and 4.5.2). Conversely, the values of v_2 and v_3 indicate optimal values for CV and UCI coherence, respectively (with higher values being preferable, typically positive, as detailed in Sections 4.5.1 and 4.5.2). A coefficient of 1 implies that the highest or lowest value (depending on the associated sign) of the specific metric will be considered, while 0 signifies that the score of the measure will be disregarded. In general, values between 0 and 1 indicate the degree of emphasis placed on a particular score.

Next, we determine the optimal value of the combined score for all normalization techniques by the following:

$$\text{Best CS} = \arg \max(CS) \quad (16)$$

That means each CS_i is a reward function (or utility function) and from its maximization, we obtain its optimal value (Pratt et al., 1995).

For our analysis, three normalized combined scores are obtained using Eq. (16). Each score is derived from the used normalization techniques and the assigned values of λ from Eq. (15). Following this, the index corresponding to the maximum score corresponds to the optimal number of topics, given by:

$$OP = \arg \max_i CS_i \quad (17)$$

In Algorithm 1, we summarize this process for determining the optimal number of topics.

In Table 4, each row presents the optimized and normalized scores for Perplexity (PP) and Coherence (UCI, UMass, CV) across all conducted experiments, as discussed in Section 4.4. These experiments use the Bag of Words (BoW) representation of encoded data, resulting in a single value for each utilized normalization technique (NT), outlined above, and contingent upon the chosen regularization values for λ .

The optimization process encompasses the normalization of both perplexity and coherence scores, amalgamating them into a unified score for each experiment. The optimal score for each normalization technique is subsequently determined by identifying the highest combined score among all conducted experiments, leading to the selection of the corresponding number of topics along with the optimal configuration. The best optimal score is then determined by identifying the highest combined score among all normalization techniques, leading to the selection of the corresponding number of topics along with the optimal configuration.

The λ values $[-1, 0, 0, 0]$ and $[1, 0, 0, 0]$ specify that only the highest or lowest perplexity score is considered respectively, among all conducted experiments for various normalization techniques, and the scores of other metrics are disregarded. Similarly, $[0, 1, 0, 0]$ and $[0, -1, 0, 0]$ focus solely on the highest and lowest UCI coherence scores, while $[0, 0, -1, 0]$ and $[0, 0, 1, 0]$ prioritize the maximum and minimum UMass coherence scores. Correspondingly, $[0, 0, 0, 1]$ and $[0, 0, 0, -1]$ correspond to the highest and lowest CV coherence scores.

For all these λ values, the combined norm score (see Eqs. (12), (13) and (14)) is calculated for each normalization technique across all conducted experiments, and the highest score signifies the specific experiment configuration, including the number of topics. The remaining rows, characterized by λ combinations between 0 and 1, indicate the degree of emphasis on a particular metric. In these rows, the majority of normalization techniques recommend 16 as the optimal number of topics (mentioned 11 times). Upon closer examination of the generated topics, it was observed that setting lambda values to $[1, 1, 1, 1]$ yielded the most intriguing topics, as illustrated in Table 4. In this scenario, the λ values emphasize achieving optimal scores for UCI and CV coherence while minimizing perplexity and UMass coherence scores. Consequently, all employed normalization techniques converge toward a consistent choice of 16 topics, with a similar configuration

suggested by 'Min-Max' and 'Z-score' (labeled as *norm 2* and *norm 3*, respectively). Then, the selection of the number of topics and configuration is determined by the normalization technique exhibiting the highest normalized combined scores, with *norm 2* being the preferred choice in this case.

Utilizing the optimal number of topics above, we show the results of this in Figs. 10 and 11. Importantly, the optimal number of topics identified for LDA is likewise applied to BERTopic, while the remaining BERTopic configurations adhere to the suggestions provided by Groo-tendorst (2022). These Figures show the application of topic modeling using both LDA and BERTopic, each model with 16 predefined topics. Figs. 10(a) and 11 (a) showcase the topic visualization for LDA and BERTopic, respectively. Figs. 10(b) and 11(b) show the four top terms associated with the initial 8 topics for both LDA and BERTopic. Furthermore, Figs. 10(c) and 11(c) depict the similarity among topics across LDA and BERTopic models.

5.4. Temporal evolution of topics

So far we considered topic modeling as a static model without the presence of time. Now we consider time as a parameter and study the evolution of topics over time.

The Latent Dirichlet Allocation (LDA) operates as a generative probabilistic model that assumes each document comprises a blend of a few topics, with each word in the document ascribed to one of these topics. LDA commonly uses a bag-of-words representation of documents, disregarding word order and context. Evaluation of the quality of generated topics in LDA relies on measures of topic coherence, assigning each document a distribution over topics and each topic a distribution over words. In contrast, BERTopic takes an embedding-based approach, leveraging contextualized word embeddings such as those provided by BERT. It incorporates contextual information and word relationships in a document, capturing at least partially semantic meaning compared to traditional bag-of-words models. BERTopic assigns topics based on dense vector representations of documents. Notably, BERTopic can be adapted for dynamic or temporal topic modeling by processing documents over time and updating the model with new information, enabling the observation of how topics evolve.

Temporal topic modeling is frequently applied in domains where understanding the evolution of topics over time is essential. This is particularly valuable in scenarios such as analyzing news articles or researching topics, where the ability to track shifting trends is crucial. In the context of sustainable energy research, it allows us to identify such trends and put these into perspective.

It is important to emphasize that LDA is innately not designed to perform temporal topic modeling. For this reason, to analyze topics over time using LDA, one would typically apply the model to different time slices independently. Still, LDA does not explicitly capture temporal dynamics, necessitating the use of supplementary techniques and manual analysis to track topic evolution.

In Fig. 12, we show the results from such a topic modeling for LDA. For this analysis, we used 16 topics. Specifically, Fig. 12(a) shows the publication count per topic over time, including the corresponding top 10 terms per topic. This figure shows a steady increase in publications over the years while not all topics increase at the same rate. Fig. 12(b) shows the hierarchical clustering of terms associated with topics throughout a dataset. It visualizes the resulting dendrogram with different colors for each cluster using complete linkage and Euclidean distance. Fig. 12(c) shows the total number of publications per topic integrated over all years. This confirms the impression from Fig. 12(a) and reveals considerable differences in research efforts. In contrast, Fig. 12(d) zooms into the results by focusing on topic #10. It shows the time evolution of all publications over time. Yet a different perspective is shown in Fig. 12(e) providing a similarity matrix of correlation values between terms that belong to topic #10.

Table 4

Optimization of the normalized scores across different configurations (see Section 4.4) for BoW. The optimal combined normalized scores for Perplexity (PP) and Coherence (UCI, UMass, CV) are determined by aggregating the normalized scores of the utilized scores.

NT	λ	Experiment No.	# of topics	Alpha	Beta	Passes	Chunksize	PP	UCI	UMass	CV	Norm_ PP	Norm_ UCI	Norm_ UMass	Norm_ CV	Combined norm score
norm 1		22	4	0.3	0.01	10	100	-33.780	0.178	-1.817	0.464	-34.629	0.189	-2.015	0.500	34.629
norm 2	[-1,0,0,0]	22	4	0.3	0.01	10	100	-33.780	0.178	-1.817	0.464	0	0.791	0.973	0.352	0
norm 3		22	4	0.3	0.01	10	100	-33.780	0.178	-1.817	0.464	-2.750	0.055	0.881	-0.930	2.750
norm 1		5	28	0.1	0.3	10	1000	-7.336	0.219	-2.716	0.542	-8.185	0.231	-2.915	0.578	-8.185
norm 2	[1,0,0,0]	5	28	0.1	0.3	10	1000	-7.336	0.219	-2.716	0.542	0	0.813	0.816	0.762	0
norm 3		5	28	0.1	0.3	10	1000	-7.336	0.219	-2.716	0.542	0.561	0.183	0.044	1.027	0.561
norm 1		21	32	1	0.3	10	1000	-7.487	0.562	-2.473	0.529	-8.336	0.573	-2.671	0.565	0.573
norm 2	[0,1,0,0]	21	32	1	0.3	10	1000	-7.487	0.562	-2.473	0.529	0.994	0	0.859	0.696	0
norm 3		21	32	1	0.3	10	1000	-7.487	0.562	-2.473	0.529	0.542	1.242	0.270	0.714	1.242
norm 1		116	32	0.3	1	10	100	-7.631	-1.274	-6.563	0.508	-8.480	-1.262	-6.761	0.544	1.262
norm 2	[0,-1,0,0]	116	32	0.3	1	10	100	-7.631	-1.274	-6.563	0.508	0.989	0	0.143	0.581	0
norm 3		116	32	0.3	1	10	100	-7.631	-1.274	-6.563	0.508	0.524	-4.429	-3.538	0.162	4.429
norm 1		17	32	0.01	1	10	100	-7.525	-0.708	-7.380	0.520	-8.374	-0.696	-7.579	0.556	7.579
norm 2	[0,0,-1,0]	17	32	0.01	1	10	100	-7.525	-0.708	-7.380	0.520	0.993	0.308	0	0.645	0
norm 3		17	32	0.01	1	10	100	-7.525	-0.708	-7.380	0.520	0.538	-2.680	-4.299	0.468	4.299
norm 1		145	4	1	0.3	10	100	-7.531	0.154	-1.664	0.458	-8.380	0.166	-1.863	0.494	-1.863
norm 2	[0,0,1,0]	145	4	1	0.3	10	100	-7.531	0.154	-1.664	0.458	0.993	0.778	0	0.320	0
norm 3		145	4	1	0.3	10	100	-7.531	0.154	-1.664	0.458	0.537	-0.017	1.023	-1.083	1.023
norm 1		119	20	0.01	1	10	100	-7.508	0.128	-3.929	0.587	-8.357	0.140	-4.127	0.623	0.623
norm 2	[0,0,0,1]	119	20	0.01	1	10	100	-7.508	0.128	-3.929	0.587	0.013	0.764	0.604	0	0
norm 3		119	20	0.01	1	10	100	-7.508	0.128	-3.929	0.587	0.540	-0.097	-1.085	2.163	2.163
norm 1		43	20	0.1	0.1	10	100	-7.888	-0.057	-3.106	0.398	-8.737	-0.046	-3.304	0.434	-0.434
norm 2	[0,0,0,-1]	43	20	0.1	0.1	10	100	-7.888	-0.057	-3.106	0.398	0.979	0.663	0.748	0	0
norm 3		43	20	0.1	0.1	10	100	-7.888	-0.057	-3.106	0.398	0.492	-0.671	-0.319	-2.611	2.611
norm 1		85	28	0.01	0.01	10	100	-33.480	-0.380	-3.744	0.405	-34.329	-0.369	-3.943	0.441	19.172
norm 2	[-0.5,0.5,-0.5,0.5]	25	16	0.3	0.01	10	100	-33.431	0.341	-2.310	0.490	0.013	0.879	0.887	0.486	0.232
norm 3		2	24	1	0.01	10	1000	-16.838	0.561	-2.389	0.546	-0.629	1.240	0.348	1.128	1.324
norm 1		97	4	0.1	0.3	10	500	-7.473	0.151	-1.699	0.454	-8.322	0.162	-1.898	0.490	-5.436
norm 2	[0.5,-0.5,0.5,-0.5]	43	20	0.1	0.1	10	100	-7.888	-0.057	-3.106	0.398	0.979	0.663	0.748	0	0.532
norm 3		43	20	0.1	0.1	10	100	-7.888	-0.057	-3.106	0.398	0.492	-0.671	-0.319	-2.611	1.728
norm 1		33	16	0.1	0.3	10	1000	-7.350	0.383	-1.952	0.519	-8.199	0.394	-2.151	0.555	-4.700
norm 2	[0.5,0.5,0.5,0.5]	30	16	1	1	10	100	-7.545	0.498	-2.134	0.560	0.992	0.965	0.918	0.861	1.868
norm 3		30	16	1	1	10	100	-7.545	0.498	-2.134	0.560	0.535	1.043	0.586	1.498	1.831
norm 1		85	28	0.01	0.01	10	100	-33.480	-0.380	-3.744	0.405	-34.329	-0.369	-3.943	0.441	19.100
norm 2	[-0.5,-0.5,-0.5,-0.5]	85	28	0.01	0.01	10	100	-33.480	-0.380	-3.744	0.405	0.011	0.487	0.636	0.039	-0.587
norm 3		85	28	0.01	0.01	10	100	-33.480	-0.380	-3.744	0.405	-2.713	-1.668	-0.914	-2.426	3.860
norm 1		85	28	0.01	0.01	10	100	-33.480	-0.380	-3.744	0.405	-34.329	-0.369	-3.943	0.441	38.345
norm 2	[-1,1,-1,1]	25	16	0.3	0.01	10	100	-33.431	0.341	-2.310	0.490	0.013	0.879	0.887	0.486	0.465
norm 3		2	24	1	0.01	10	1000	-16.838	0.561	-2.389	0.546	-0.629	1.240	0.348	1.128	2.648
norm 1		97	4	0.1	0.3	10	500	-7.473	0.151	-1.699	0.454	-8.322	0.162	-1.898	0.490	-10.872
norm 2	[1,-1,1,-1]	43	20	0.1	0.1	10	100	-7.888	-0.057	-3.106	0.398	0.979	0.663	0.748	0	1.064
norm 3		43	20	0.1	0.1	10	100	-7.888	-0.057	-3.106	0.398	0.492	-0.671	-0.319	-2.611	3.455
norm 1		85	28	0.01	0.01	10	100	-33.480	-0.380	-3.744	0.405	-34.329	-0.369	-3.943	0.441	38.200
norm 2	[-1,-1,-1,-1]	85	28	0.01	0.01	10	100	-33.480	-0.380	-3.744	0.405	0.011	0.487	0.636	0.039	-1.173
norm 3		85	28	0.01	0.01	10	100	-33.480	-0.380	-3.744	0.405	-2.713	-1.668	-0.914	-2.426	7.721
norm 1		33	16	0.1	0.3	10	1000	-7.350	0.383	-1.952	0.519	-8.199	0.394	-2.151	0.555	-9.400
norm 2	[1,1,1,1]	30	16	1	1	10	100	-7.545	0.498	-2.134	0.560	0.992	0.965	0.918	0.861	3.735
norm 3		30	16	1	1	10	100	-7.545	0.498	-2.134	0.560	0.535	1.043	0.586	1.498	3.661

Next, we conduct a temporal topic modeling analysis for BERTopic. This examination aims to assess whether its capability to capture contextual information and adapt to new data makes it more suitable for dynamic topic modeling. The continuous updating of the model with new documents over time enables a more seamless and integrated observation of the dynamic changes and evolution of topics. The results of this analysis are shown in Fig. 13.

Fig. 13(a) depicts the temporal evolution of the 16 explicitly defined topics during BERTopic training. Fig. 13(b) focuses on the analysis of term frequency for topic #1 across time. Additionally, Fig. 13(c) illustrates the hierarchical clustering of terms linked with the topics.

When examining Figs. 12(a) and 13(a), it becomes evident that the LDA model delves into a spectrum of topics encompassing energy, renewable sources, efficiency, environmental impact, urban management, carbon emissions, economic variables, sustainability policies, and production processes. Conversely, the BERTopic model explores topics including energy systems, materials, and devices pertinent to energy storage and conversion, catalysis, power regulation systems, magnetic characteristics, coastal and oceanic occurrences, optical attributes, network communication, and electrolytes. Notably, both models touch upon common themes such as energy, materials, catalysis, power systems, and optical properties, albeit with different emphases

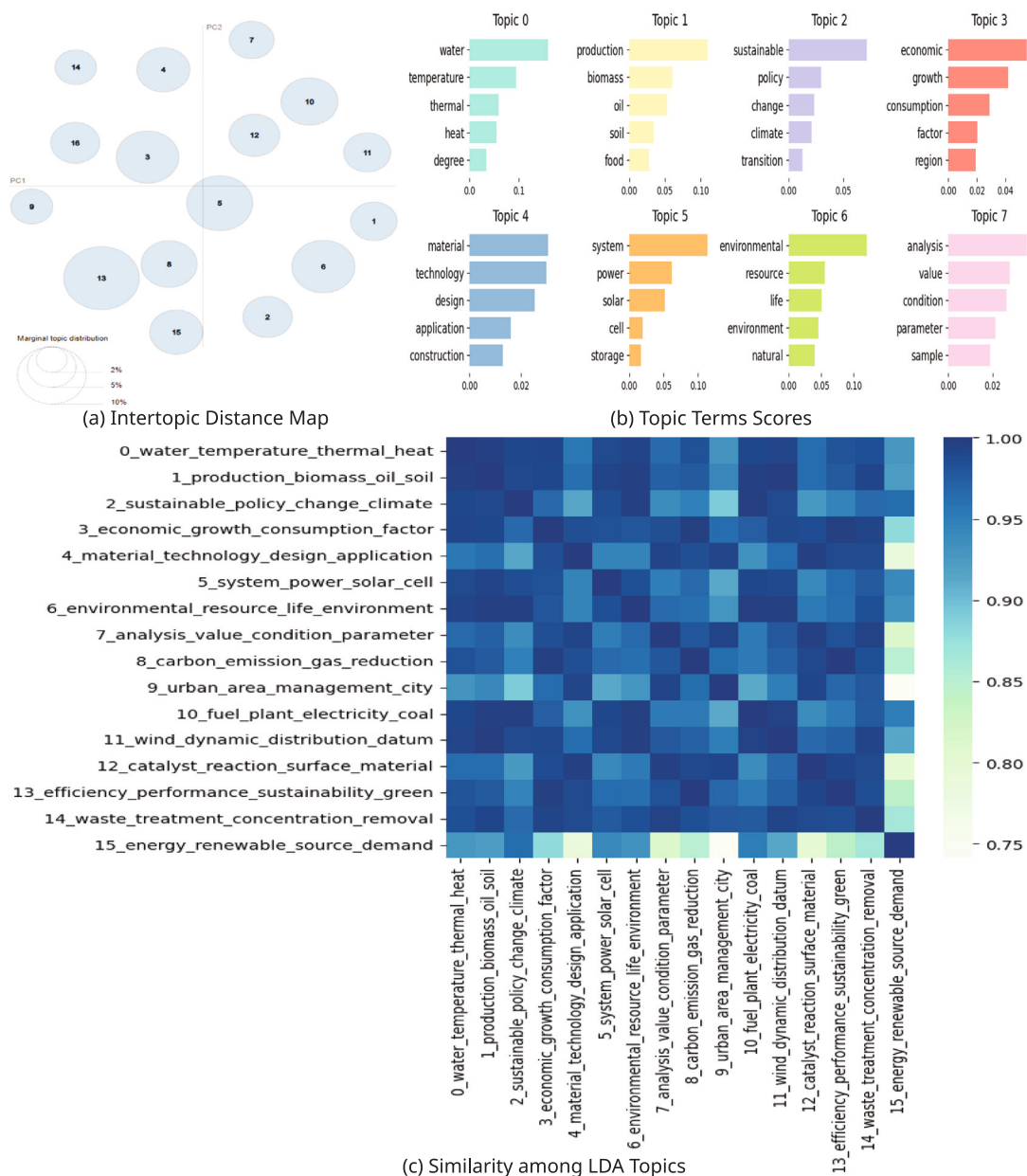


Fig. 10. Topic modeling using LDA with 16 predefined topics. (a) Corresponds to the LDA model, showcasing its topics. (b) Displays the top four terms for the first 8 topics. (c) Explores the correlation similarity between topics.

and granularity. This similarity further emphasizes that employing 16 topics seems to be an appropriate selection for both models.

6. Discussion

In the literature, one can find numerous techniques for topic modeling, each presenting distinct advantages and disadvantages. In Table 5, we present an overview of the most popular methods used within this domain. Despite the prevalent belief that topic modeling uncovers meaningful and valuable latent concepts, validating this assumption is challenging due to the unsupervised nature of topic discovery. Therefore, the absence of a definitive ground truth regarding the optimal number of topics and their composition underscores the importance of integrating external metrics to evaluate the latent spaces generated by topic models.

A reason that makes it difficult to determine the optimal number of topics is that it invariably depends on various factors. One of these

is the characteristics of the textual data (corpora) itself. For instance, consider the following questions: Do the documents cover a broad spectrum of terms, ideas, subjects, and themes? Do the documents in the corpus consist of meticulously crafted academic paper summaries on specific, narrow topics? Or, could the documents be random abstracts lacking any clear unifying theme or structure? The determination of the optimal number of topics is intricately tied to such contextual factors, and there is no one-size-fits-all answer.

In this study, we compare LDA and BERTopic representing traditional and modern topic modeling approaches. While LDA functions as a probabilistic generative model, assuming documents comprise distinct topics and assigning words to specific topics based on probabilities, BERTopic utilizes transformer-based models to create context-aware embeddings for text documents. Potentially, BERTopic is less directly interpretable than LDA, because BERTopic emphasizes contextually rich topics. Furthermore, it can be computationally demanding, especially with large datasets, yet optimizations like UMAP can manage

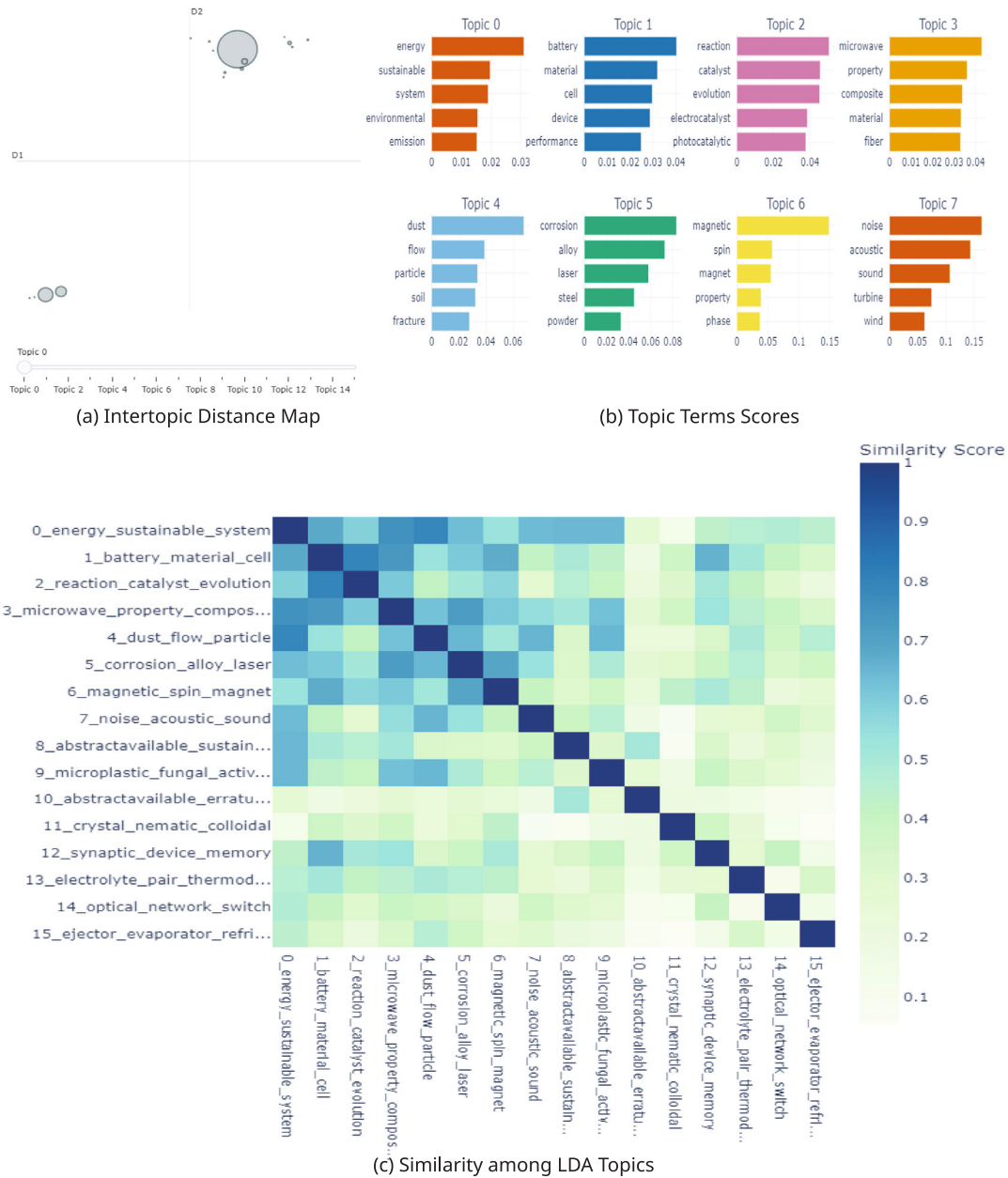


Fig. 11. Topic modeling using BERTopic with 16 predefined topics is depicted. (a) Corresponds to the BERTopic model, showcasing its topics. (b) Displays the top four terms for the first 8 topics. (c) Explores the correlation similarity between topics.

its scalability. Importantly, both topic modeling techniques have in common that they are sensitive to hyperparameters but can effectively handle varying document lengths and multiple topics per document.

A fundamental feature of topic models like LDA or BERTopic is their capability to dynamically select the number of topics to be modeled or require a specified number during the training phase. However, determining the optimal number of topics is not a straightforward process. Notably, when both LDA and BERTopic were allowed to autonomously determine their respective number of topics, a very large number of topics was generated, namely 100 topics for LDA and 1101 topics for BERTopic, see Section 5.1.

This dynamic topic selection feature in both LDA and the BERTopic transformer allows for flexibility in associating specific terms with different topics. This capability enables the creation of topics that highlight terms with lower frequency in documents. For example, the term “nuclear” appears in eight different topics among the 1101 topics in BERTopic, yet it does not rank among the top ten terms in any of

the 16 selected topics. Similarly, when the number of topics is set to 16, the term “solar” is exclusively present in one topic. However, when the number of topics is not specified, it is distributed across 19 topics out of the 1101 in BERTopic.

It is essential to highlight that both LDA and BERTopic models exhibit stochastic behavior, meaning that each execution of these models may produce slightly different results. Thus, in this evaluation aimed at establishing the ideal number of topics for the LDA model, multiple models were run for each chosen number of topics, assessed by a set of scores. A total of 150 experiments were conducted, covering various predefined parameters including the number of topics for each data encoding type (such as WoB and TF-IDF).

To guide our experiments, we employed a random search approach within a predetermined parameter range. The specific configuration is detailed in Table 2 and Figs. 5 and 6 are outlining results for different parameter values for alpha, beta, passes, and chunk size.

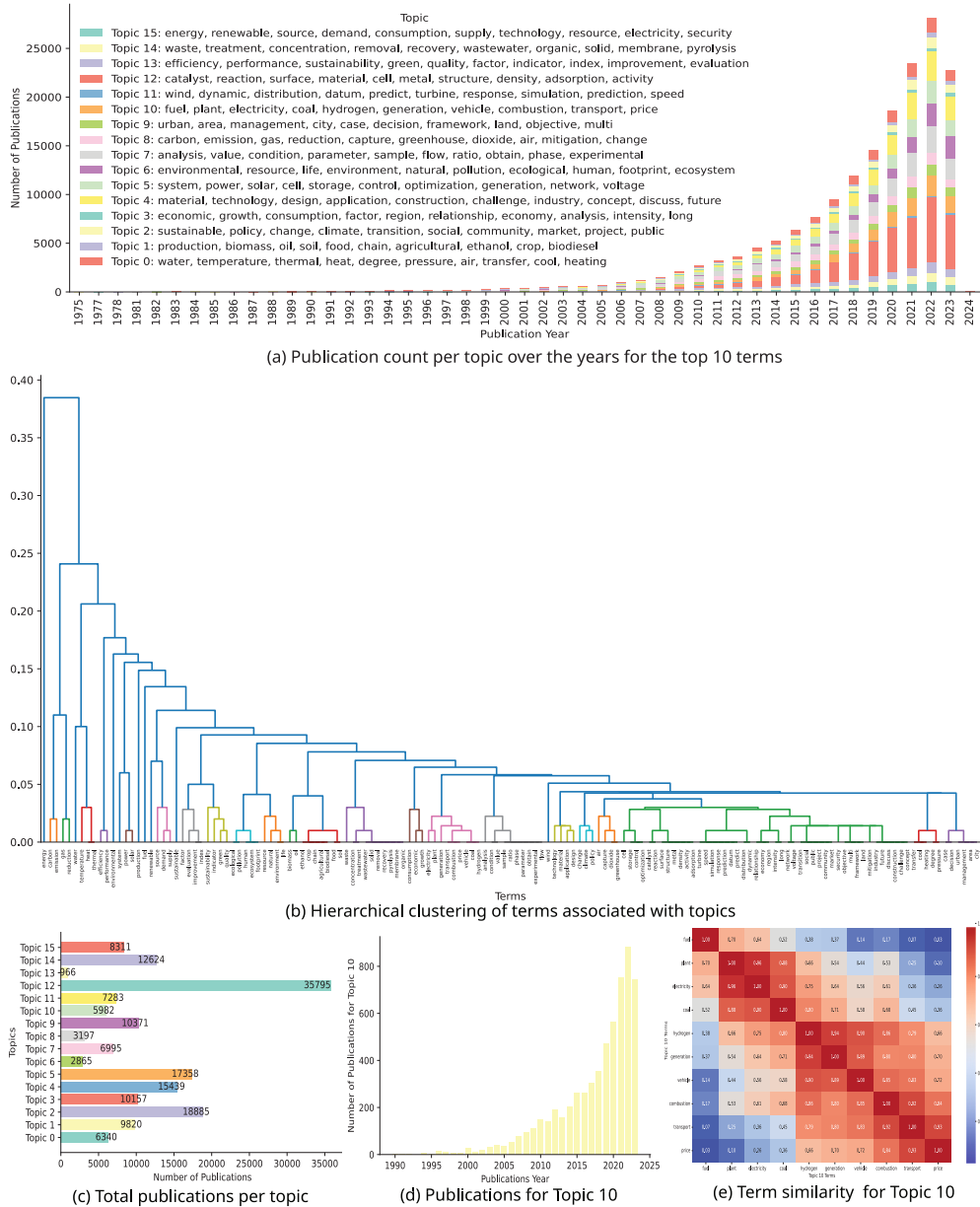


Fig. 12. Summary of results of the time evolution for LDA with 16 topics. (a) Topic distributions over years. (b) Hierarchical clustering of terms associated with topics. (c) Total publications per topic. (d) Publications for topic 10. (e) Term similarity for topic 10.

The motivation for combining various evaluation scores to obtain an overall measure is based on the reported weaknesses of the individual scores (Röder et al., 2015; Doogan and Buntine, 2021). For instance, perplexity, while widely used, lacks consistency with human interpretability (Chang et al., 2009). Similarly, the performance of different coherence scores varies across different models and corpora in terms of interpretability (Röder et al., 2015; Doogan and Buntine, 2021). By combining and optimizing these various measures, one can aim for a more comprehensive evaluation of topic model quality and interpretability. Additionally, this approach enables the automatic identification of the optimal number of topics in topic modeling without requiring manual intervention.

To address this issue, we introduced a mathematical optimization based on a utility function that incorporates multiple normalization techniques and a regularization parameter (λ). This parameter assigns weights to specific scores, with the goal of determining the most suitable number of topics across all the conducted experiments (see

Section 5.3). Significantly, the optimal number of topics identified for LDA models was subsequently applied to BERTopic.

When studying the optimization of parameters (see Table 4), we find that reducing the number of topics to a smaller values makes the clustering process more rigid, highlighting only the topics with the most dominant terms. In this scenario, terms are less likely to appear across multiple topics, resulting in minimal overlap. Conversely, having a larger number of topics results in softer clusters with significant overlap between them. This can also be demonstrated by observing the maximum perplexity scores when the lambda values are set to $[-1, 0, 0, 0]$, as depicted in the first rows of Table 4, where the number of topics is 4. Conversely, when the perplexity score is minimized by lambda setting to $[1, 0, 0, 0]$, the corresponding number of topics is notably larger at 28. This can also be ascribed to the predetermined values of alpha and beta, as outlined in Section 4.4.

It is important to note that LDA is not designed to perform temporal topic modeling. To analyze topics over time using LDA, one would typically apply the model to different time slices independently.

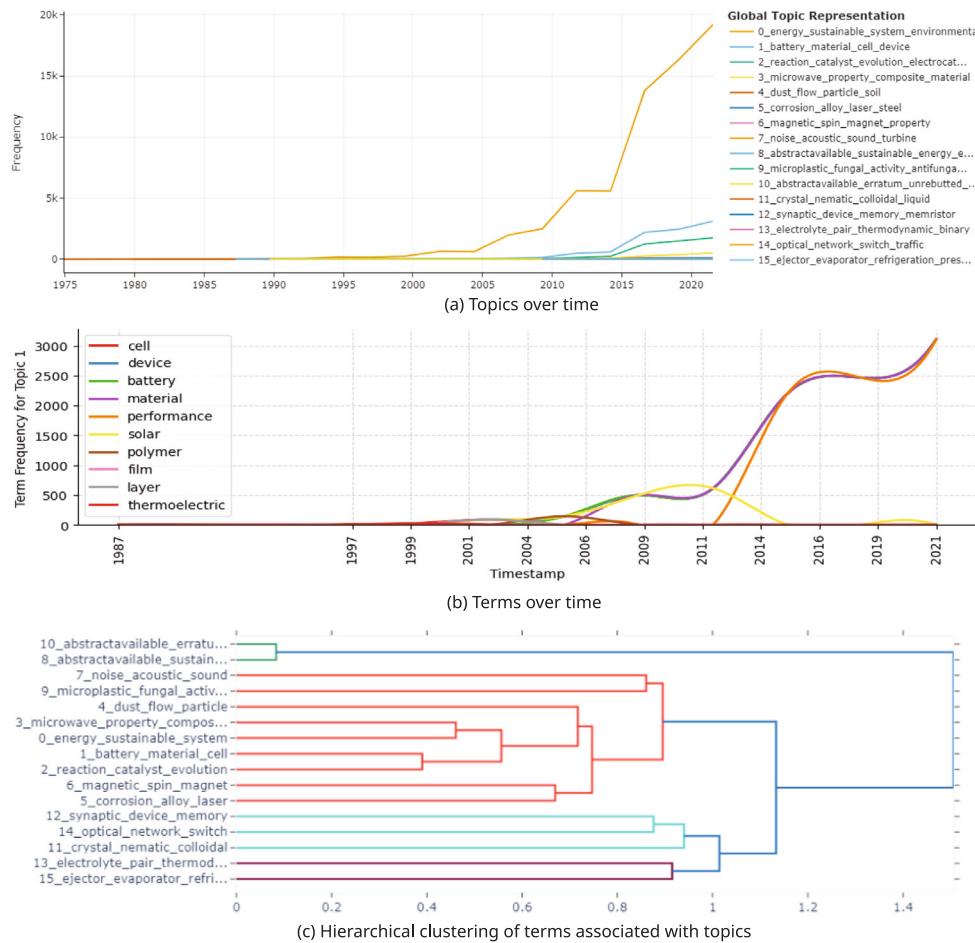


Fig. 13. Subfigure (a) illustrates the temporal distribution of the 16 topics explicitly outlined during BERTopic training, highlighting the term frequency analysis for topic #1 across time, as demonstrated in Subfigure (b). Subfigure (c) displays the hierarchical clustering of terms linked with the topics.

Table 5 Overview of topic modeling techniques. Highlighted features include Interpretability, Scalability, Handling of varying document lengths, Sensitivity to hyperparameters, and Handling of multiple topics per document.				
Model	Interpretability	Scalability	Handling diverse document lengths	Sensitivity to hyperparameter
LDA	Good, interpretable topics represented by words	Relatively scalable	Effective	Sensitive
NMF	Highly interpretable with non-negative weights	Scalable	Effective	Less sensitive
PLSA	Good interpretability with word probabilities	Less scalable	Effective	Sensitive
CTM	Good interpretability, models topic correlations	Less scalable	Effective	Sensitive
DTM	Interpretability with temporal evolution	Moderately scalable	Effective with temporal aspects	Sensitive to temporal parameters
hLDA	Interpretable with hierarchical structure	Less scalable	Effective	Sensitive
BERTopic	Contextually rich but may be less interpretable	Computationally intensive	Effective with transformer embeddings	Sensitive to clustering parameters

Nevertheless, LDA does not explicitly capture temporal dynamics, necessitating the use of supplementary techniques and manual analysis to track topic evolution. Dynamic Topic Models (DTM) could offer a solution, but its computational demands can be significant, particularly for large datasets with numerous time slices, as they require estimating topic distributions for each time period.

Nevertheless, it is crucial to highlight that we have not extensively delved into BERTopic fine-tuning or thoroughly explored its clustering techniques. Our primary focus has been on determining the optimal number of topics in LDA model, emphasizing diverse metrics scores that indicate variations across different topic numbers. In response to this, we have proposed an optimization technique based on various normalization methods. As a result, the optimal number of topics for LDA

model has been applied to BERTopic. While conducting experiments with BERTopic's hyperparameters or clustering techniques may yield promising results, there is a lack of a straightforward justification for determining the optimal number of topics.

7. Conclusion

Topic modeling plays a critical role in organizing vast amounts of textual data, enabling improved information retrieval, content summarization, and contextual understanding. As data continues to grow in complexity and volume, topic modeling remains an indispensable tool for harnessing meaningful insights from unstructured text, making it an exciting area of research and application in the realm of natural language processing, machine learning and data science.

In our study, we investigated both Latent Dirichlet Allocation (LDA) and BERTopic for topic modeling. We utilized LDA, a probabilistic generative model, within our optimization framework to identify underlying topics within the data. Our approach is based on normalized perplexity and coherence scores to determine the optimal number of topics. Insights derived from LDA's topics guided the selection of the optimal number of topics for BERTopic. BERTopic, which harnesses pre-trained language models, was then employed to achieve a more nuanced understanding of the topics and to capture contextually semantic relationships that might not be as readily discernible with LDA.

Regarding results for publication about sustainable energy research, our findings reveal that reducing the number of topics results in more rigid clusters, emphasizing dominant terms, while a larger number leads to softer clusters with a notable overlap. This observation is attributed to predetermined values and optimization choices made during the study.

CRediT authorship contribution statement

Amer Farea: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Shailesh Tripathi:** Writing – review & editing, Validation, Methodology. **Galina Glazko:** Writing – review & editing, Validation, Methodology. **Frank Emmert-Streib:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

ST thanks the Austrian Federal Ministry of Digital and Economic Affairs, the National Foundation for Research, Technology and Development, and the Christian Doppler Research Association for financial support.

References

- Al Moubayed, N., Wall, D., McGough, A.S., 2017. Identifying changes in the cybersecurity threat landscape using the LDA-web topic modelling data search engine. In: *Human Aspects of Information Security, Privacy and Trust: 5th International Conference, HAS 2017, Held as Part of HCI International 2017, Vancouver, BC, Canada, July 9–14, 2017, Proceedings 5*. Springer, pp. 287–295.
- Alagheband, M.R., Mashatan, A., Zihayat, M., 2020. Time-based gap analysis of cybersecurity trends in academic and digital media. *ACM Trans. Manag. Inf. Syst. (TMIS)* 11 (4), 1–20.
- Alghamdi, R., Alfalqi, K., 2015. A survey of topic modeling in text mining. *Int. J. Adv. Comput. Sci. Appl. (IJACSA)* 6 (1).
- Amouzgar, M., Glass, D.R., Baskar, R., Averbukh, I., Kimmey, S.C., Tsai, A.G., Hartmann, F.J., Bendall, S.C., 2022. Supervised dimensionality reduction for exploration of single-cell data by HSS-LDA. *Patterns* 3 (8).
- Angelov, D., 2020. Top2vec: Distributed representations of topics. *arXiv preprint arXiv:2008.09470*.
- Arun, R., Suresh, V., Veni Madhavan, C., Narasimha Murthy, M., 2010. On finding the natural number of topics with latent dirichlet allocation: Some observations. In: *Advances in Knowledge Discovery and Data Mining: 14th Pacific-Asia Conference, PAKDD 2010, Hyderabad, India, June 21–24, 2010. Proceedings. Part 1* 14. Springer, pp. 391–402.
- Bassiou, N., Kotropoulos, C., 2011. RPLSA: A novel updating scheme for probabilistic latent semantic analysis. *Comput. Speech Lang.* 25 (4), 741–760.
- Bianchi, F., Terragni, S., Hovy, D., 2020a. Pre-training is a hot topic: Contextualized document embeddings improve topic coherence. *arXiv preprint arXiv:2004.03974*.
- Bianchi, F., Terragni, S., Hovy, D., Nozza, D., Fersini, E., 2020b. Cross-lingual contextualized topic models with zero-shot learning. *arXiv preprint arXiv:2004.07737*.
- Blei, D.M., Lafferty, J.D., 2006. Dynamic topic models. In: *Proceedings of the 23rd International Conference on Machine Learning*. pp. 113–120.
- Blei, D.M., Lafferty, J.D., 2007. A correlated topic model of science.
- Blei, D.M., Ng, A.Y., Jordan, M.I., 2003. Latent dirichlet allocation. *J. Mach. Learn. Res.* 3 (Jan), 993–1022.
- Bosch, A., Zisserman, A., Munoz, X., 2006. Scene classification via pLSA. In: *Computer Vision–ECCV 2006: 9th European Conference on Computer Vision, Graz, Austria, May 7–13, 2006, Proceedings, Part IV* 9. Springer, pp. 517–530.
- Bouma, G., 2009. Normalized (pointwise) mutual information in collocation extraction. *Proc. GSCCL* 30, 31–40.
- Cao, Z., Li, S., Liu, Y., Li, W., Ji, H., 2015. A novel neural topic model and its supervised extension. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 29.
- Cao, J., Xia, T., Li, J., Zhang, Y., Tang, S., 2009. A density-based method for adaptive LDA model selection. *Neurocomputing* 72 (7–9), 1775–1781.
- Chandran, N.V., Anoop, V., Asharaf, S., 2023. Topicstriker: A topic kernels-powered approach for text classification. *Res. Eng.* 17, 100949.
- Chang, J., Gerrish, S., Wang, C., Boyd-Graber, J., Blei, D., 2009. Reading tea leaves: How humans interpret topic models. *Adv. Neural Inf. Process. Syst.* 22.
- del Gobbo, E., Guarino, A., Cafarelli, B., Grilli, L., Limone, P., 2023. Automatic evaluation of open-ended questions for online learning. A systematic mapping. *Stud. Educ. Eval.* 77, 101258.
- Deveaud, R., SanJuan, E., Bellot, P., 2014. Accurate and effective latent concept modeling for ad hoc information retrieval. *Doc. Numér.* 17 (1), 61–84.
- Devlin, J., Chang, M.-W., Lee, K., Toutanova, K., 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- Dieng, A.B., Ruiz, F.J., Blei, D.M., 2020. Topic modeling in embedding spaces. *Trans. Assoc. Comput. Linguist.* 8, 439–453.
- Doogan, C., Buntine, W., 2021. Topic model or topic twaddle? Re-evaluating demand interpretability measures. In: *North American Association for Computational Linguistics 2021. Association for Computational Linguistics (ACL)*, pp. 3824–3848.
- Dumais, S., et al., 1994. Latent semantic indexing (LSI) and TREC-2. In: *Nist Special Publication Sp. National Institute of Standards & Technology*, p. 105.
- Edara, D.C., Vanukuri, L.P., Sistla, V., Kolli, V.K.K., 2023. Sentiment analysis and text categorization of cancer medical records with LSTM. *J. Ambient Intell. Humaniz. Comput.* 14 (5), 5309–5325.
- Eligüzel, N., Cetinkaya, C., Dereli, T., 2022. A novel approach for text categorization by applying hybrid genetic bat algorithm through feature extraction and feature selection methods. *Expert Syst. Appl.* 202, 117433.
- Emmert-Streib, F., Dehmer, M., 2019. Understanding statistical hypothesis testing: The logic of statistical inference. *Mach. Learn. Knowl. Extr.* 1 (3), 945–962.
- Emmert-Streib, F., Yang, Z., Feng, H., Tripathi, S., Dehmer, M., 2020. An introductory review of deep learning for prediction models with big data. *Front. Artif. Intell.* 3, 4.
- Garg, M., Rangra, P., 2022. Bibliometric analysis of latent Dirichlet allocation. *DESIDOC J. Libr. Inf. Technol.* 42 (2).
- Griffiths, T., Jordan, M., Tenenbaum, J., Blei, D., 2003. Hierarchical topic models and the nested Chinese restaurant process. *Adv. Neural Inf. Process. Syst.* 16.
- Griffiths, T.L., Steyvers, M., 2004. Finding scientific topics. *Proc. Natl. Acad. Sci.* 101 (suppl_1), 5228–5235.

- Grootendorst, M., 2022. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*.
- Gupta, R.K., Agarwalla, R., Naik, B.H., Evuri, J.R., Thapa, A., Singh, T.D., 2022. Prediction of research trends using LDA based topic modeling. *Glob. Transitions Proc.* 3 (1), 298–304.
- Gurcan, F., Cagiltay, N.E., 2022. Exploratory analysis of topic interests and their evolution in bioinformatics research using semantic text mining and probabilistic topic modeling. *IEEE Access* 10, 31480–31493.
- Hall, D., Jurafsky, D., Manning, C.D., 2008. Studying the history of ideas using topic models. In: *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*. pp. 363–371.
- Hammarde, K.A., Proenca, H., 2023. Information retrieval: Recent advances and beyond. *arXiv preprint arXiv:2301.08801*.
- He, Y., Jiang, F., Chu, X., Li, P., 2022. Automated Chinese essay scoring from multiple traits. In: *Proceedings of the 29th International Conference on Computational Linguistics*. pp. 3007–3016.
- Ho, K.T., Bui, Q.V., Bui, M., 2018. Dynamic social network analysis using author-topic model. In: *International Conference on Innovations for Community Services*. Springer, pp. 47–62.
- Hofmann, T., 1999. Probabilistic latent semantic indexing. In: *Proceedings of the 22nd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*. pp. 50–57.
- Hofmann, T., 2001. Unsupervised learning by probabilistic latent semantic analysis. *Mach. Learn.* 42, 177–196.
- Hong, L., Davison, B.D., 2010. Empirical study of topic modeling in twitter. In: *Proceedings of the First Workshop on Social Media Analytics*. pp. 80–88.
- Islam, T., 2019. Yoga-veganism: Correlation mining of twitter health data. *arXiv preprint arXiv:1906.07668*.
- Joachims, T., et al., 1997. A probabilistic analysis of the Rocchio algorithm with TFIDF for text categorization. In: *ICML. Vol. 97*, Citeseer, pp. 143–151.
- Jockers, M.L., Mimmo, D., 2013. Significant themes in 19th-century literature. *Poetics* 41 (6), 750–769.
- Kakkonen, T., Myller, N., Sutinen, E., Timonen, J., 2008. Comparison of dimension reduction methods for automated essay grading. *J. Educ. Technol. Soc.* 11 (3), 275–288.
- Kawai, M., Sato, H., Shiohama, T., 2022. Topic model-based recommender systems and their applications to cold-start problems. *Expert Syst. Appl.* 202, 117129.
- Klakow, D., Peters, J., 2002. Testing the correlation of word error rate and perplexity. *Speech Commun.* 38 (1–2), 19–28.
- Kushwaha, A.K., Kar, A.K., Roy, S.K., Ilavarasan, P.V., 2022. Capricious opinions: A study of polarization of social media groups. *Gov. Inf. Q.* 39 (3), 101709.
- Lan, Z., Chen, M., Goodman, S., Gimpel, K., Sharma, P., Soricut, R., 2019. Albert: A lite bert for self-supervised learning of language representations. *arXiv preprint arXiv:1909.11942*.
- Landauer, T.K., Dumais, S.T., 1997. A solution to Plato's problem: The latent semantic analysis theory of acquisition, induction, and representation of knowledge. *Psychol. Rev.* 104 (2), 211.
- Landauer, T.K., Foltz, P.W., Laham, D., 1998. An introduction to latent semantic analysis. *Discourse Process.* 25 (2–3), 259–284.
- Larochelle, H., Lauly, S., 2012. A neural autoregressive topic model. *Adv. Neural Inf. Process. Syst.* 25.
- Lau, J.H., Newman, D., Baldwin, T., 2014. Machine reading tea leaves: Automatically evaluating topic coherence and topic model quality. In: *Proceedings of the 14th Conference of the European Chapter of the Association for Computational Linguistics*. pp. 530–539.
- Le, Q., Mikolov, T., 2014. Distributed representations of sentences and documents. In: *International Conference on Machine Learning*. PMLR, pp. 1188–1196.
- Lee, D.D., Seung, H.S., 1999. Learning the parts of objects by non-negative matrix factorization. *Nature* 401 (6755), 788–791.
- Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C.H., Kang, J., 2020. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 36 (4), 1234–1240.
- Li, N., Guo, B., Liu, Y., Yao, L., Liu, J., Yu, Z., 2022. AskMe: joint individual-level and community-level behavior interaction for question recommendation. *World Wide Web* 1–24.
- Li, H., Qian, Y., Jiang, Y., Liu, Y., Zhou, F., 2023. A novel label-based multimodal topic model for social media analysis. *Decis. Support Syst.* 164, 113863.
- Liu, Y., Liu, Z., Chua, T.-S., Sun, M., 2015. Topical word embeddings. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 29.
- Liu, Y., Niculescu-Mizil, A., Gryc, W., 2009. Topic-link LDA: joint models of topic and author community. In: *Proceedings of the 26th Annual International Conference on Machine Learning*. pp. 665–672.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V., 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*.
- Maier, D., Waldherr, A., Miltner, P., Wiedemann, G., Niekler, A., Keinert, A., Pfetsch, B., Heyer, G., Reber, U., Häussler, T., et al., 2021. Applying LDA topic modeling in communication research: Toward a valid and reliable methodology. In: *Computational Methods for Communication Science*. Routledge, pp. 13–38.
- Maurya, S.K., Singh, D., Maurya, A.K., 2023. Deceptive opinion spam detection approaches: a literature survey. *Appl. Intell.* 53 (2), 2189–2234.
- McCallum, A., Wang, X., Corrada-Emmanuel, A., 2007. Topic and role discovery in social networks with experiments on enron and academic email. *J. Artif. Intell. Res.* 30, 249–272.
- McInnes, L., Healy, J., 2017. Accelerated hierarchical density based clustering. In: *2017 IEEE International Conference on Data Mining Workshops. ICDMW, IEEE*, pp. 33–42.
- Michel, J.-B., Shen, Y.K., Aiden, A.P., Veres, A., Gray, M.K., Team, G.B., Pickett, J.P., Hoiberg, D., Clancy, D., Norvig, P., et al., 2011. Quantitative analysis of culture using millions of digitized books. *Science* 331 (6014), 176–182.
- Nallapati, R.M., Ahmed, A., Xing, E.P., Cohen, W.W., 2008. Joint latent topic models for text and citations. In: *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 542–550.
- Nguyen, D.Q., Billingsley, R., Du, L., Johnson, M., 2015. Improving topic models with latent feature word representations. *Trans. Assoc. Comput. Linguist.* 3, 299–313.
- Porteous, I., Newman, D., Ihler, A., Asuncion, A., Smyth, P., Welling, M., 2008. Fast collapsed gibbs sampling for latent dirichlet allocation. In: *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 569–577.
- Pratt, J.W., Raiffa, H., Schlaifer, R., 1995. *Introduction to Statistical Decision Theory*. MIT Press.
- PubMed, 2013. PubMed. URL: <https://pubmed.ncbi.nlm.nih.gov/>.
- Qiang, J., Chen, P., Wang, T., Wu, X., 2017. Topic modeling over short texts by incorporating word embeddings. In: *Advances in Knowledge Discovery and Data Mining: 21st Pacific-Asia Conference, PAKDD 2017, Jeju, South Korea, May 23–26, 2017, Proceedings, Part II 21*. Springer, pp. 363–374.
- Ramathulasi, T., Rajasekharababu, M., 2022. Augmented latent Dirichlet allocation model via word embedded clusters for mashup service clustering. *Concurr. Comput.: Pract. Exper.* 34 (15), e6896.
- Rao, R.S., Pais, A.R., 2019. Detection of phishing websites using an efficient feature-based machine learning framework. *Neural Comput. Appl.* 31, 3851–3873.
- Reimers, N., Gurevych, I., 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*.
- Röder, M., Both, A., Hinneburg, A., 2015. Exploring the space of topic coherence measures. In: *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining*. pp. 399–408.
- Romberg, S., Horster, E., Lienhart, R., 2009. Multimodal pLSA on visual features and tags. In: *2009 IEEE International Conference on Multimedia and Expo. IEEE*, pp. 414–417.
- Sbalchiero, S., 2018. Topic detection: a statistical model and a quali-quantitative method. In: *Tracing the Life Cycle of Ideas in the Humanities and Social Sciences*. Springer, pp. 189–210.
- Sbalchiero, S., Eder, M., 2020. Topic modeling, long texts and the best number of topics. Some problems and solutions. *Qual. Quant.* 54, 1095–1108.
- Scopus, 2013. Scopus. URL: <https://www.scopus.com/>.
- Shafiei, M.M., Milios, E.E., 2006. Latent Dirichlet co-clustering. In: *Sixth International Conference on Data Mining. ICDM'06, IEEE*, pp. 542–551.
- Shamna, P., Govindan, V., Nazeer, K.A., 2022. Content-based medical image retrieval by spatial matching of visual words. *J. King Saud Univ.-Comput. Inf. Sci.* 34 (2), 58–71.
- Sharma, C., Sharma, S., Sakshi, 2022. Latent DIRICHLET allocation (LDA) based information modelling on BLOCKCHAIN technology: a review of trends and research patterns used in integration. *Multimedia Tools Appl.* 81 (25), 36805–36831.
- Shi, M., Liu, J., Zhou, D., Tang, M., Cao, B., 2017. WE-LDA: a word embeddings augmented LDA model for web services clustering. In: *2017 IEEE International Conference on Web Services. IcwS, IEEE*, pp. 9–16.
- Sia, S., Dalmia, A., Mielke, S.J., 2020. Tired of topic models? clusters of pretrained word embeddings make for fast and good topics too!. *arXiv preprint arXiv:2004.14914*.
- Srinivasarao, U., Sharaff, A., 2022. Email thread sentiment sequence identification using PLSA clustering algorithm. *Expert Syst. Appl.* 193, 116475.
- Srivastava, A.N., Sahami, M., 2009. *Text Mining: Classification, Clustering, and Applications*. CRC Press.
- Steyvers, M., Griffiths, T., 2007. Probabilistic topic models. In: *Handbook of Latent Semantic Analysis*. Vol. 427, pp. 424–440.
- Sufi, F., 2023. A new AI-based semantic cyber intelligence agent. *Future Internet* 15 (7), 231.
- Syed, S., Spruit, M., 2017. Full-text or abstract? examining topic coherence scores using latent dirichlet allocation. In: *2017 IEEE International Conference on Data Science and Advanced Analytics. DSAA, IEEE*, pp. 165–174.
- Tong, Z., Zhang, H., 2016. A text mining research based on LDA topic modelling. In: *International Conference on Computer Science, Engineering and Information Technology*. pp. 201–210.
- Wang, W., Guo, B., Shen, Y., Yang, H., Chen, Y., Suo, X., 2021. Robust supervised topic models under label noise. *Mach. Learn.* 110, 907–931.
- Watanabe, S., Iwata, T., Hori, T., Sako, A., Ariki, Y., 2011. Topic tracking language model for speech recognition. *Comput. Speech Lang.* 25 (2), 440–461.

- Webber, W., Moffat, A., Zobel, J., 2010. A similarity measure for indefinite rankings. *ACM Trans. Inf. Syst. (TOIS)* 28 (4), 1–38.
- Wei, X., Croft, W.B., 2006. LDA-based document models for ad-hoc retrieval. In: *Proceedings of the 29th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*. pp. 178–185.
- WoS, 2013. *Web of science*. URL: <https://www.webofscience.com/wos/>.
- Wu, M., Long, R., Chen, F., Chen, H., Bai, Y., Cheng, K., Huang, H., 2023. Spatio-temporal difference analysis in climate change topics and sentiment orientation: Based on LDA and BiLSTM model. *Resour. Conserv. Recy.* 188, 106697.
- Wu, H., Wang, Y., Cheng, X., 2008. Incremental probabilistic latent semantic analysis for automatic question recommendation. In: *Proceedings of the 2008 ACM Conference on Recommender Systems*. pp. 99–106.
- Yu, D., Xiang, B., 2023. Discovering topics and trends in the field of artificial intelligence: Using LDA topic modeling. *Expert Syst. Appl.* 120114.
- Zhang, D.C., Lauw, H.W., 2022. Variational graph author topic modeling. In: *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. pp. 2429–2438.
- Zhao, H., Phung, D., Huynh, V., Jin, Y., Du, L., Buntine, W., 2021. Topic modelling meets deep neural networks: A survey. *arXiv preprint arXiv:2103.00498*.